

Introduction to Computer Vision

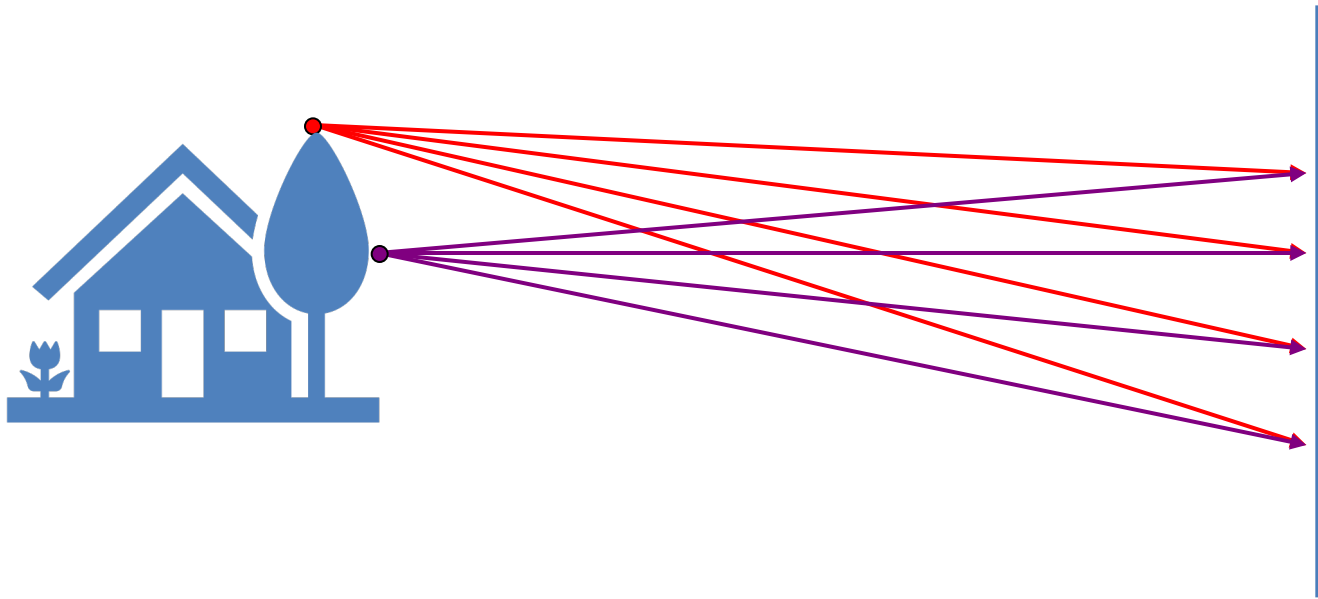


Lec.02: Image formation
2021.09.22

Today

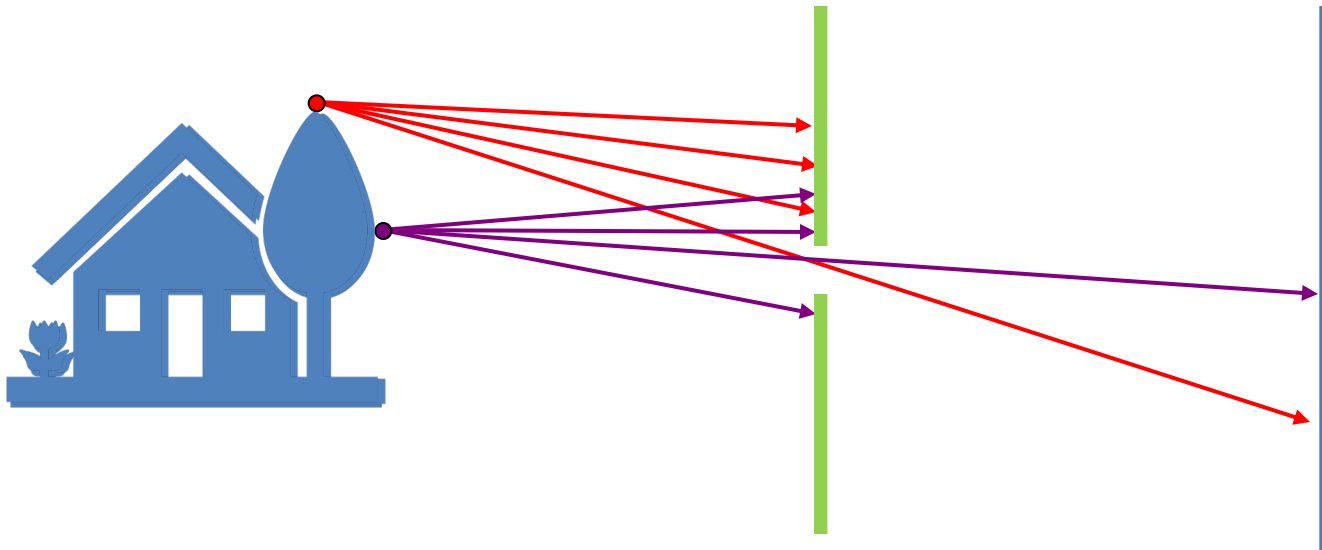
- **Camera and lens.**
- Geometric image formation.
- Photometric image formation.

Image formation



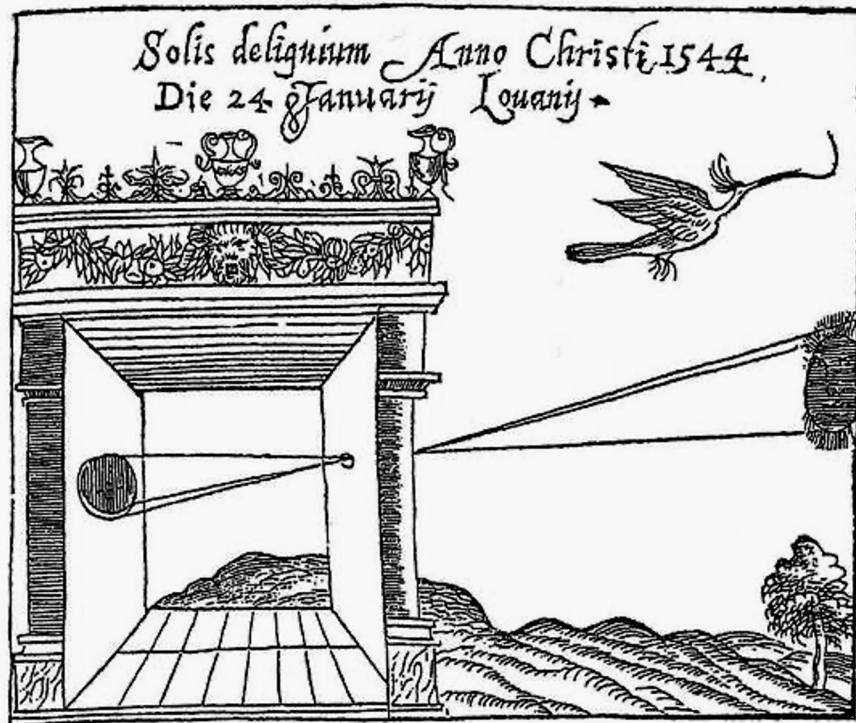
- Let's design a camera
 - Idea 1: put a piece of film in front of an object
 - Do we get a reasonable image?
 - No. This is a bad camera (not one-to-one).

Pinhole camera



- Add a barrier to block off most of the rays
 - The opening known as the **aperture**

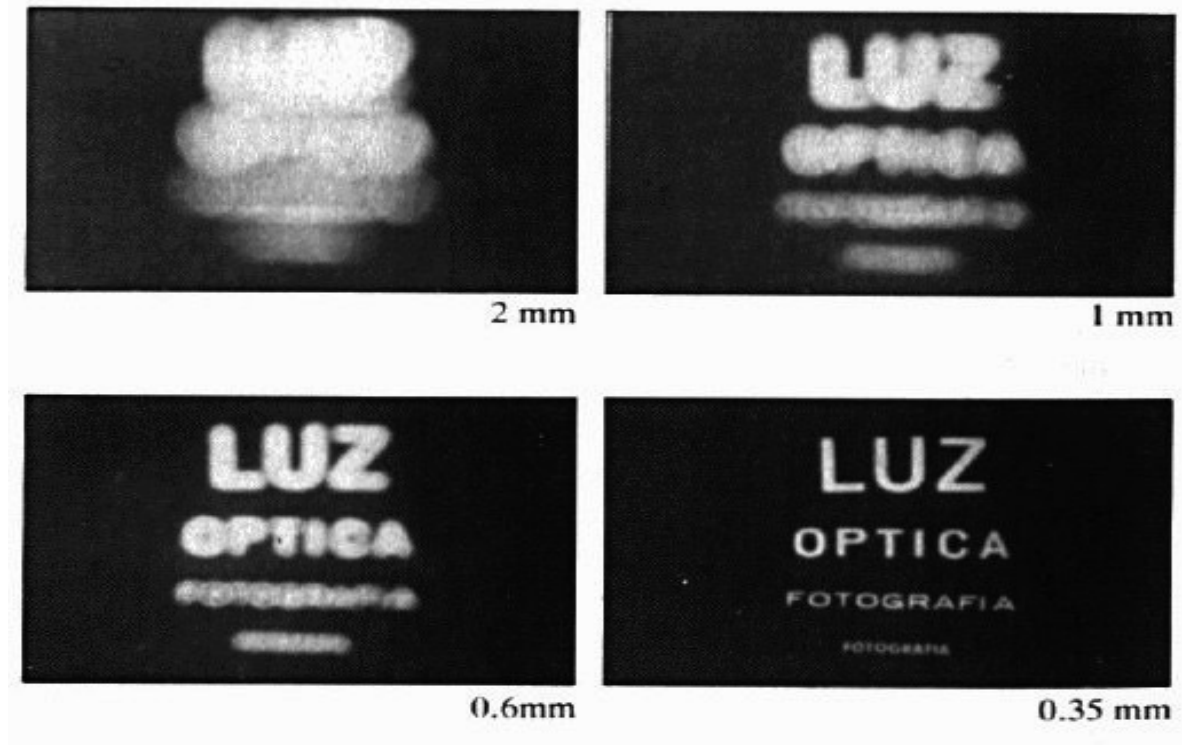
Pinhole camera



Gemma Frisius, 1558

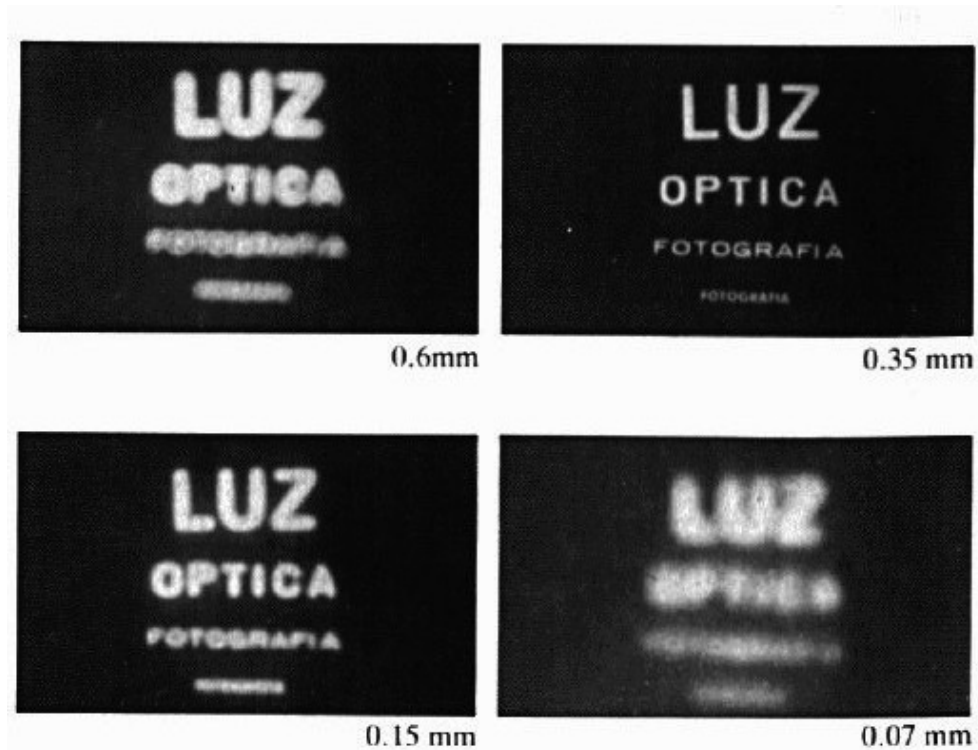
- Basic principle known to Mozi (470-390 BC), Aristotle (384-322 BC)

Shrinking the aperture



- Why not make the aperture as small as possible?

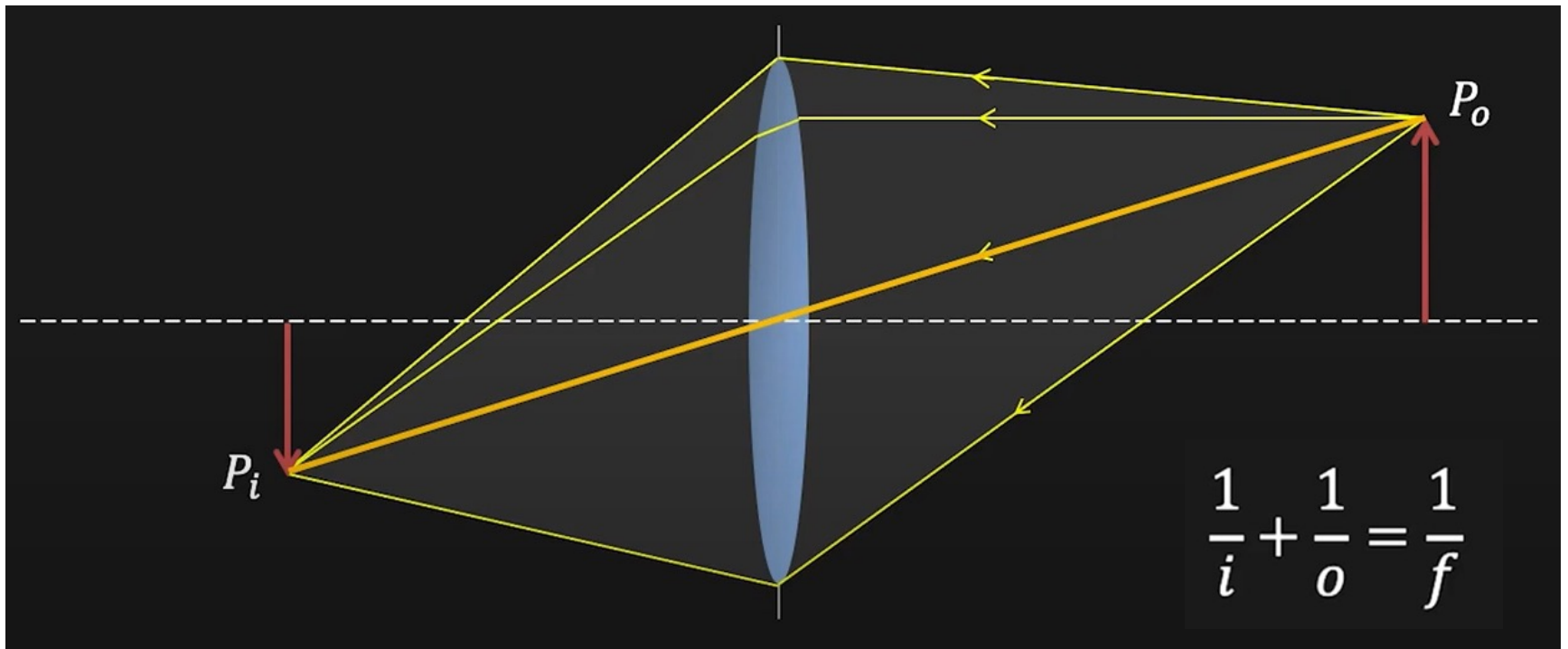
Shrinking the aperture



- Why not make the aperture as small as possible?
 - Less light gets through
 - *Diffraction* effects...

Lens

Same projection as pinhole, but gather more light.



Focal length

$$\frac{1}{i} + \frac{1}{o} = \frac{1}{f} \quad \Rightarrow \quad \text{If } o = \infty, \text{ then } f = i$$

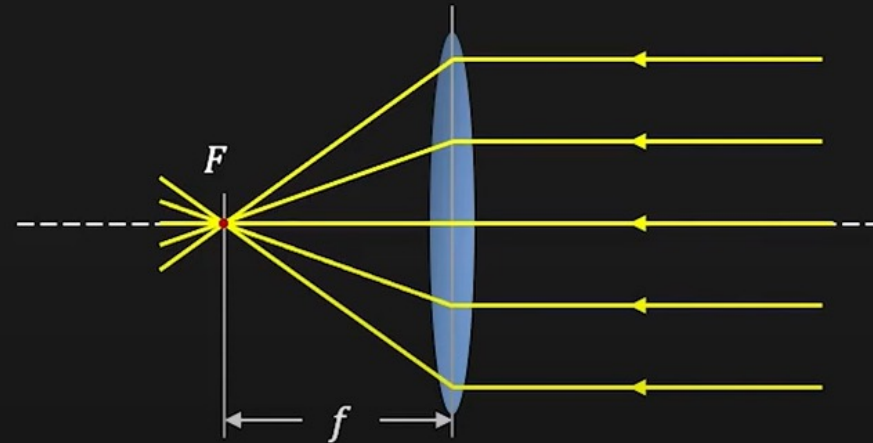
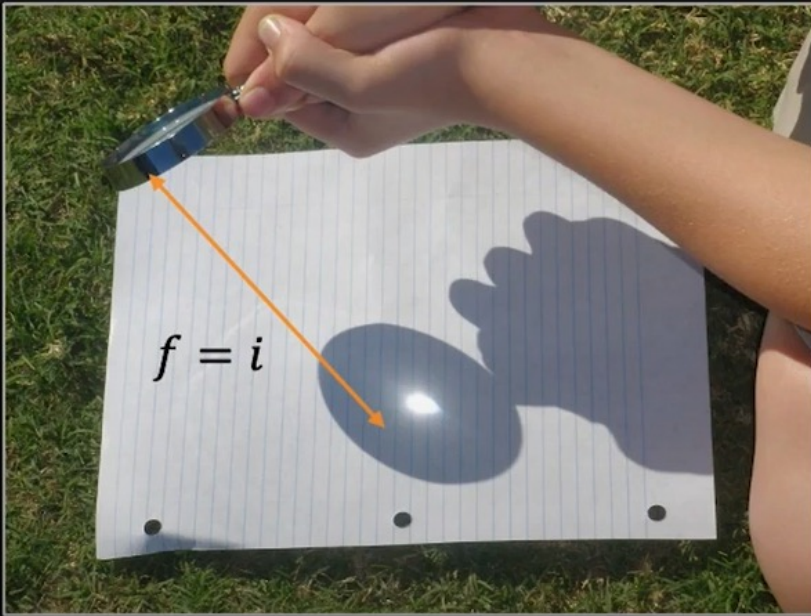
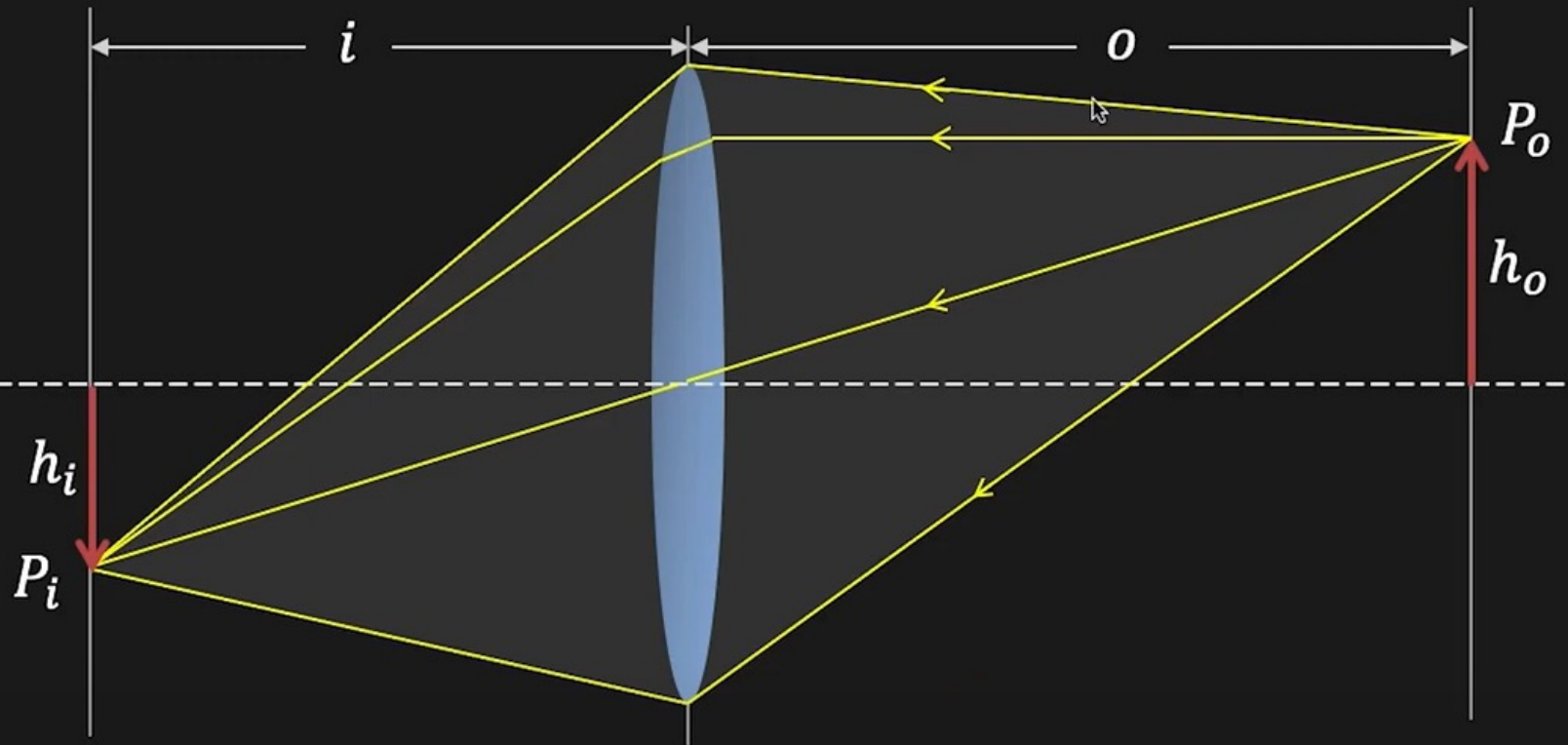


Image Magnification



Magnification: $m = \frac{h_i}{h_o}$

Image Magnification

- Changing focal length



24mm



50mm



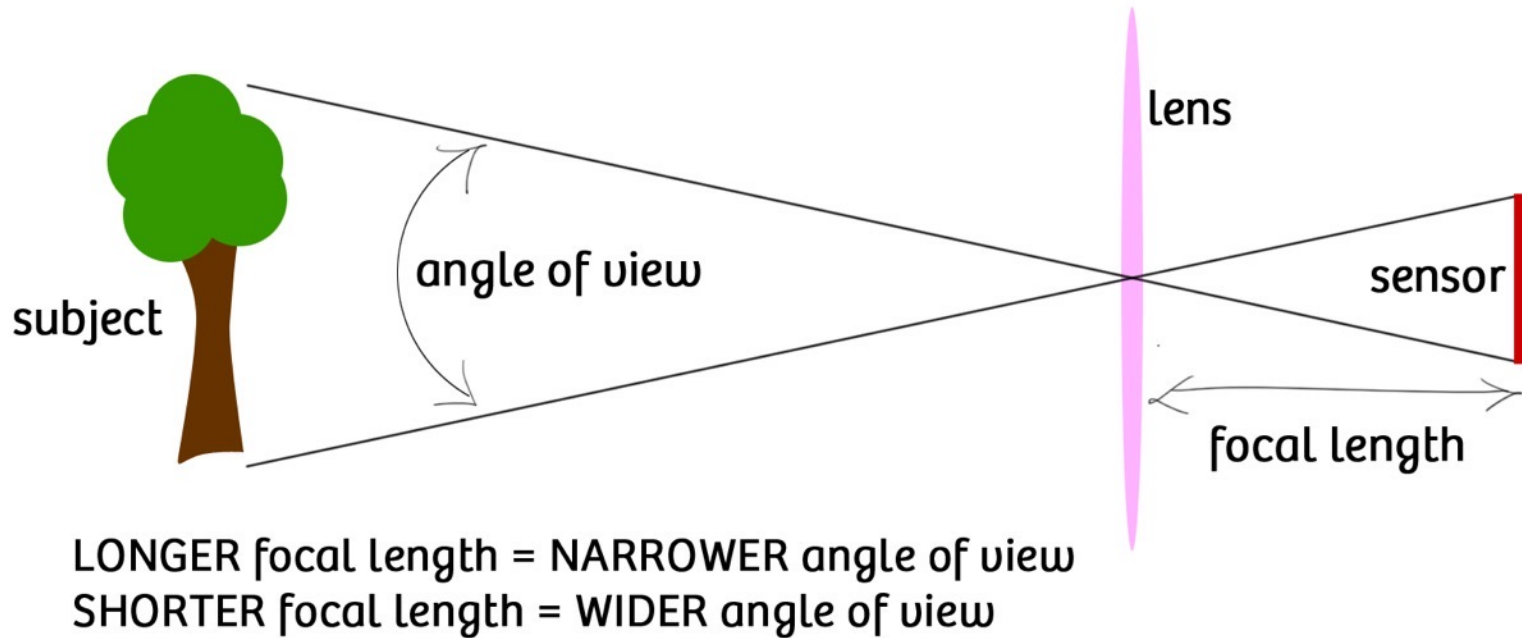
200mm



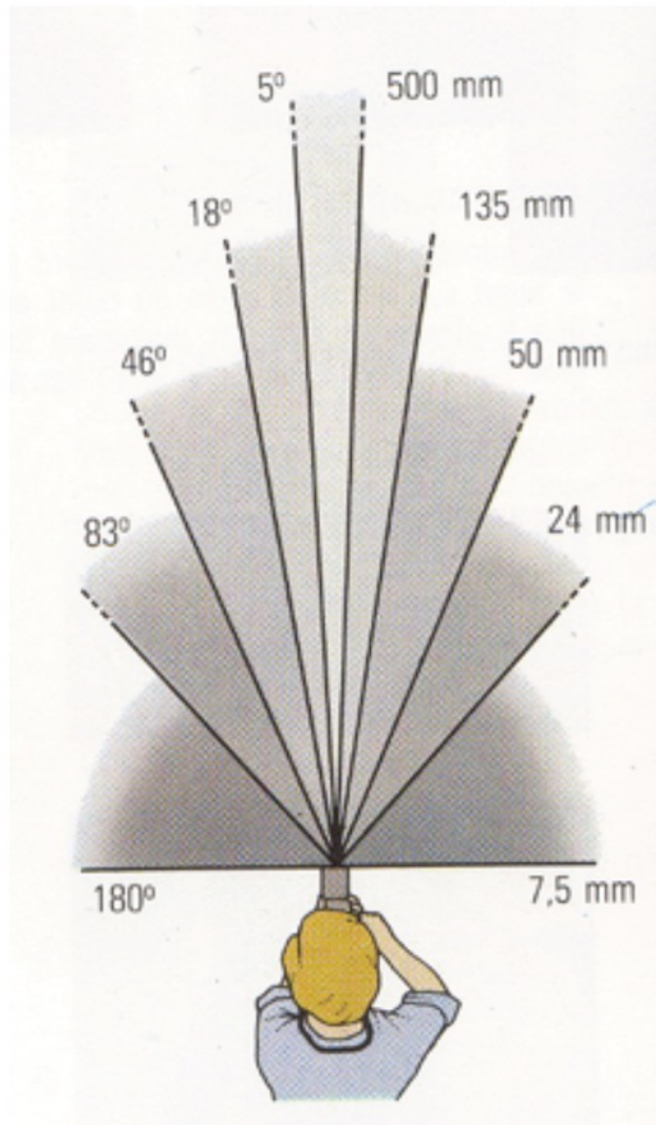
800mm



Field of View (FOV)



Focal length in practice



24mm



50mm

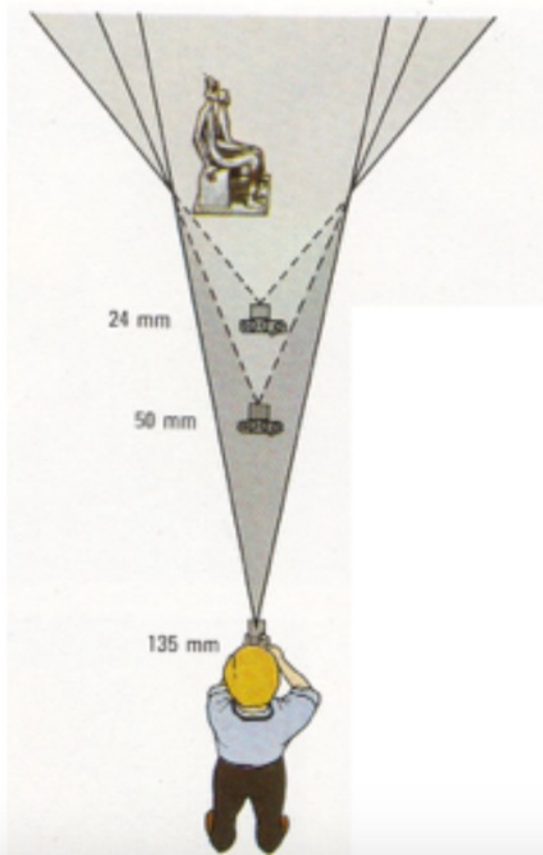


135mm



Focal length vs. viewpoint

- **Telephoto makes it easier to select background (a small change in viewpoint is a big change in background).**



Grand-angle 24 mm



Normal 50 mm



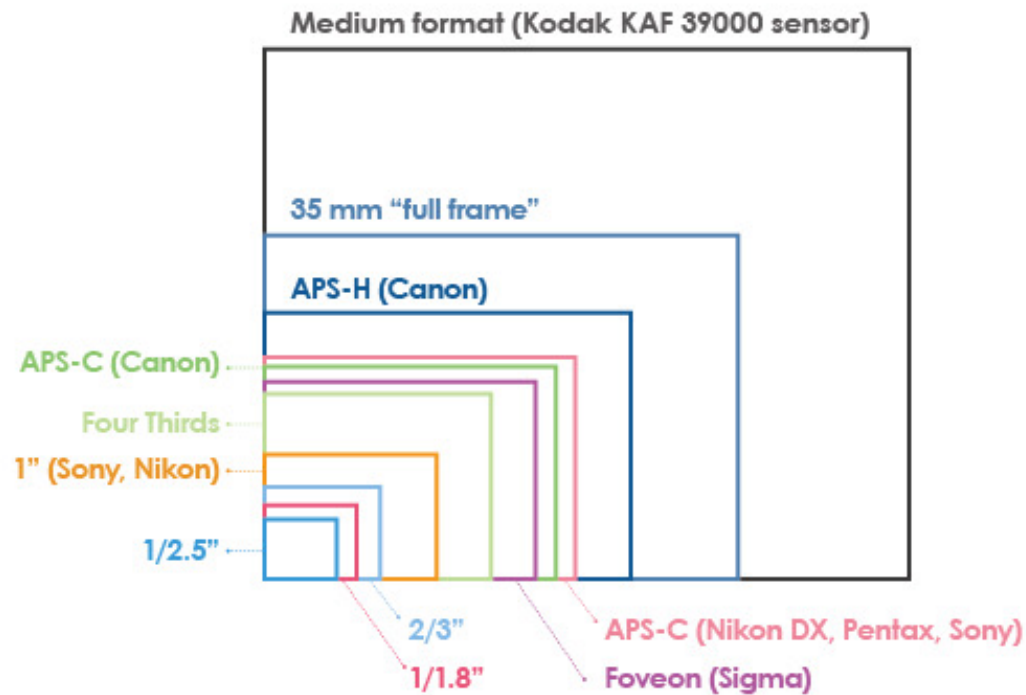
Longue focale 135 mm



Fredo Durand

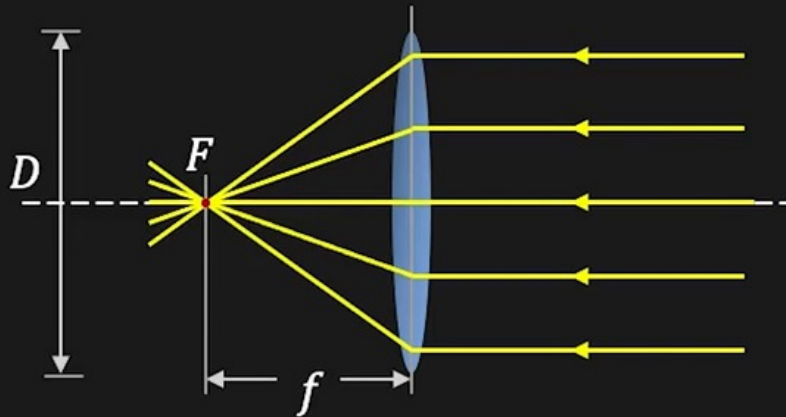
Sensor size

- FOV also depends on sensor size

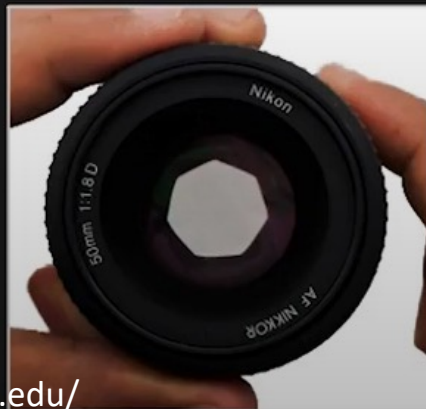
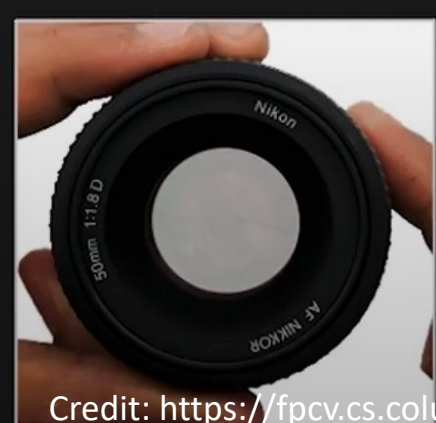


Aperture

Light receiving area of lens, indicated by lens diameter.



Aperture can be reduced/increased to control image brightness



F-number

- More convenient to represent aperture as a fraction of focal length: $D = \frac{f}{N}$
- F-Number: $N = \frac{f}{D}$
- Example: For a 50 mm focal length, $N=1.8$ ($D=27.8$ mm) when aperture is fully open.



$N = 1.8$



$N = 4$

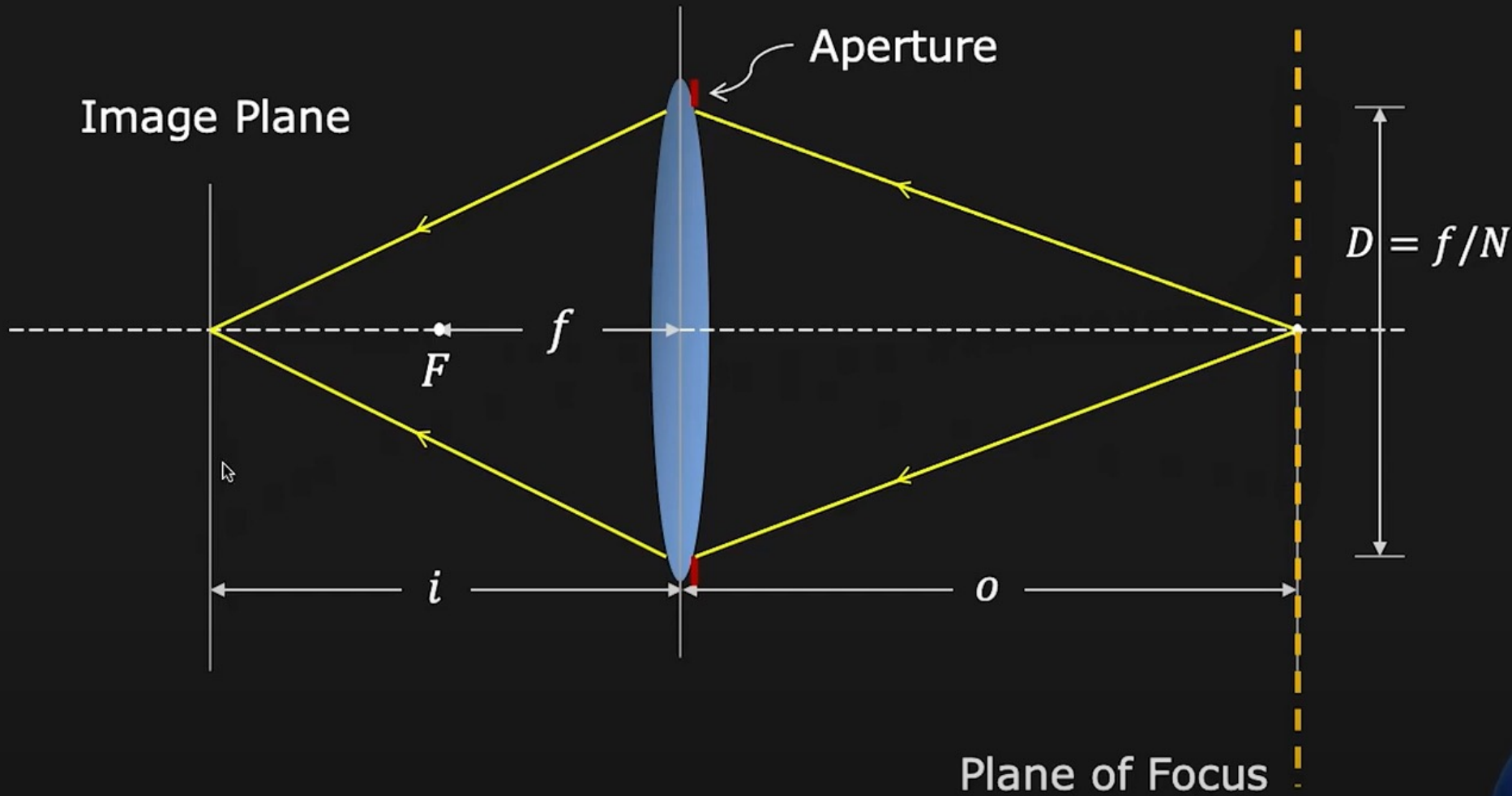


$N = 8$



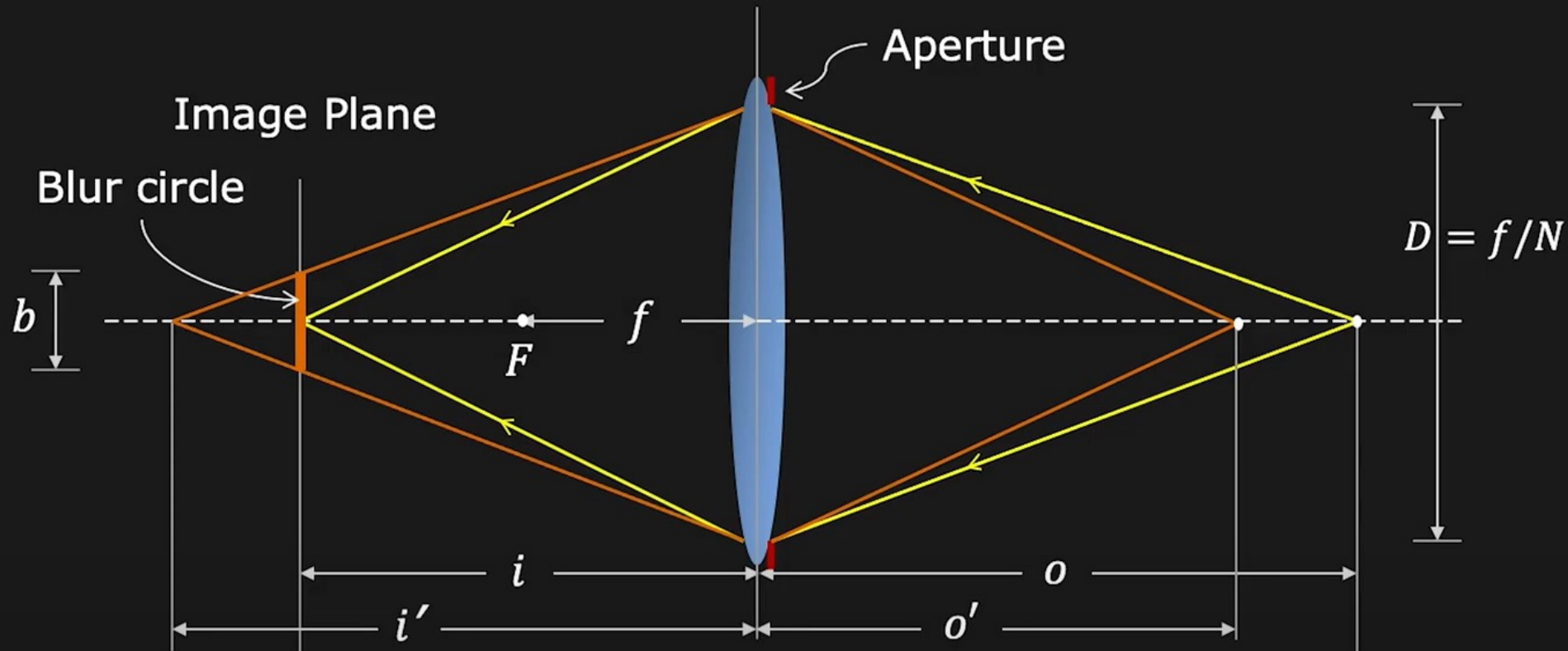
$N = 11$

Lens Defocus



Credit: <https://fpcv.cs.columbia.edu/>

Lens Defocus



Credit: <https://fpcv.cs.columbia.edu/>

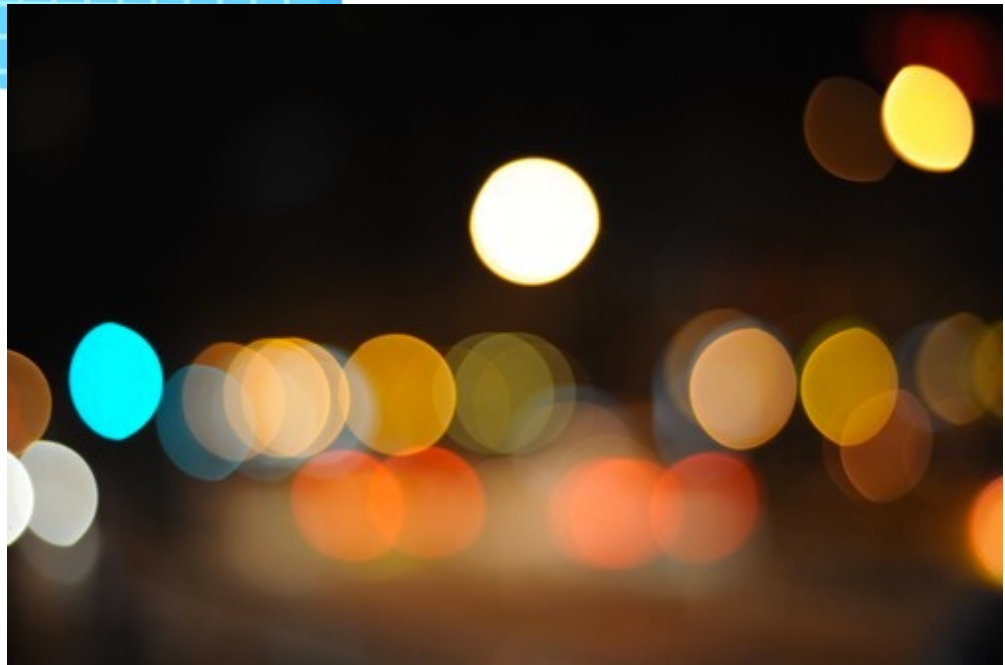
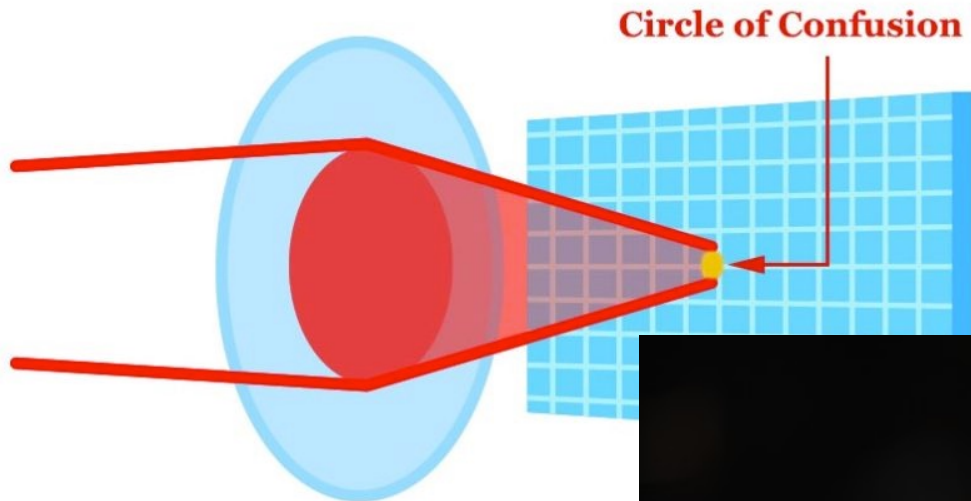
From similar triangles:

$$\frac{b}{D} = \frac{|i' - i|}{i'},$$

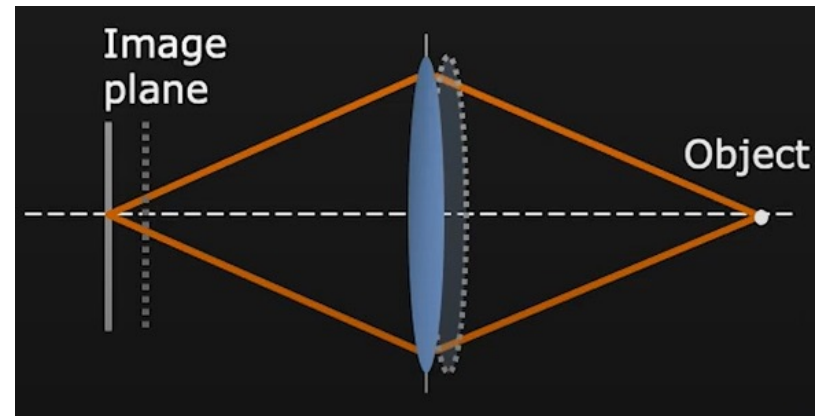
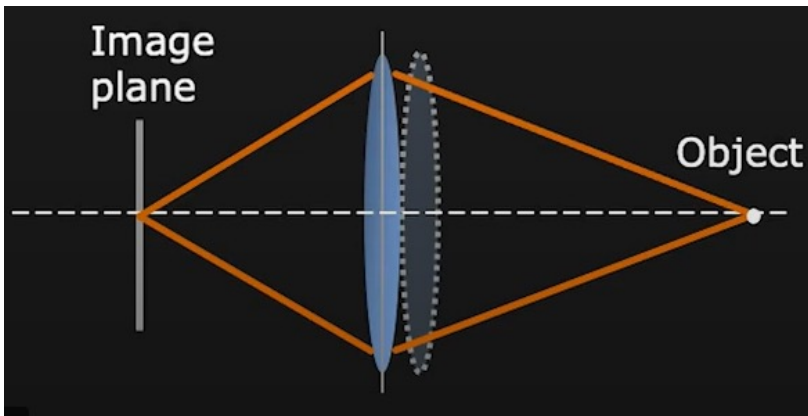
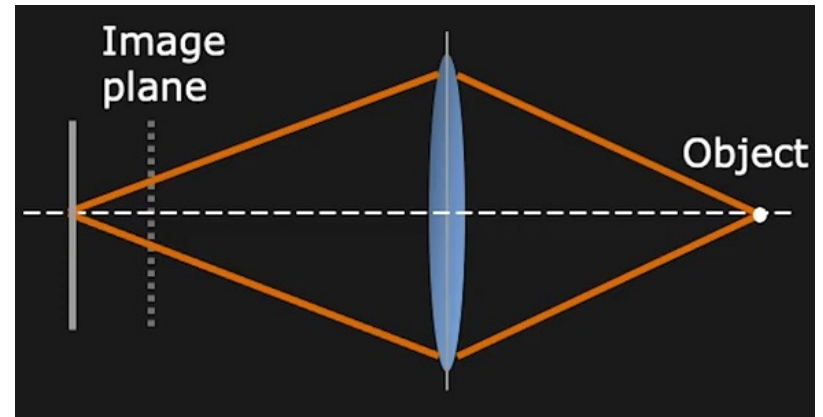
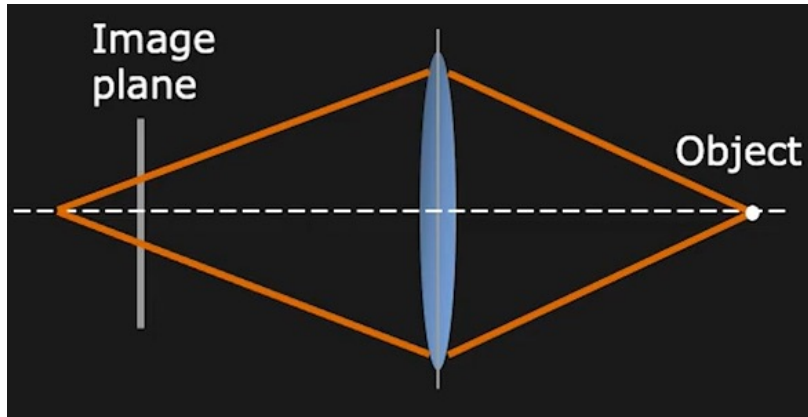
Blur circle diameter:

$$b = \frac{D}{i'} |i' - i|, b \propto D \propto \frac{1}{N}$$

Lens Defocus



Focusing



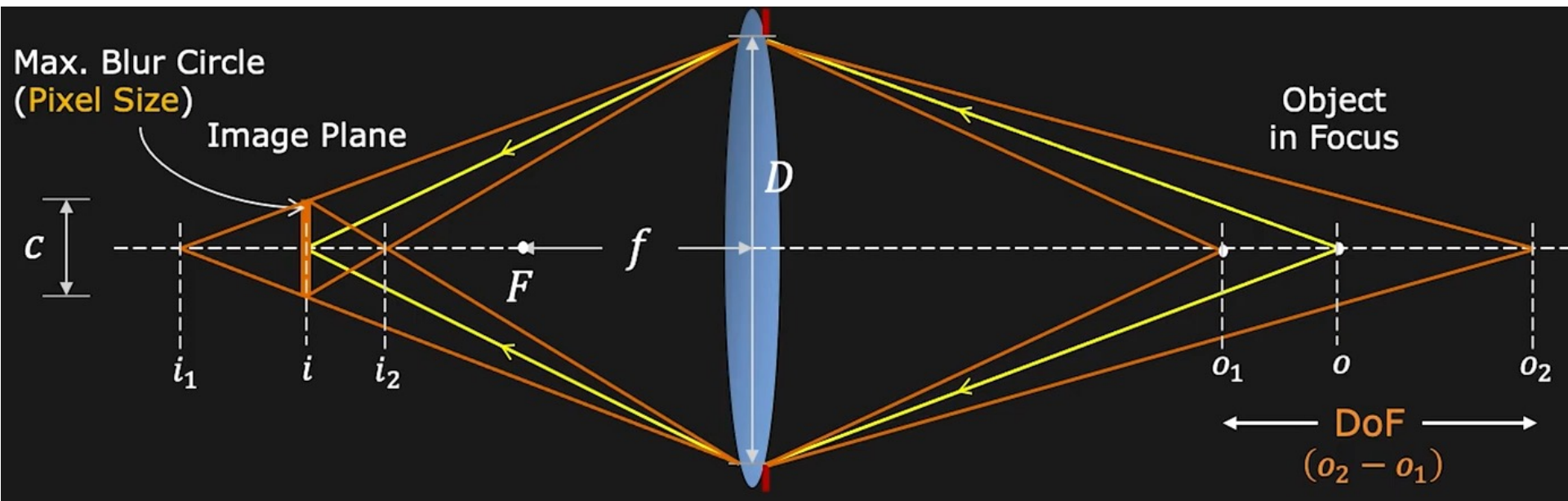
Credit: <https://fpcv.cs.columbia.edu/>

Depth of Field



- Range of object distances over which the image is sufficiently well focused, i.e. range over which blur b is less than pixel size.

Depth of Field (DoF)



Credit: <https://fpcv.cs.columbia.edu/>

- $c = \frac{f^2(o - o_1)}{No_1(o - f)}, c = \frac{f^2(o_2 - o)}{No_2(o - f)}$
- Depth of Field: $o_2 - o_1 = \frac{2of^2cN(o - f)}{f^4 - c^2N^2(o - f)^2}$

How to blur the background

- Large aperture
- Long focal length
- Near foreground
- Far Background

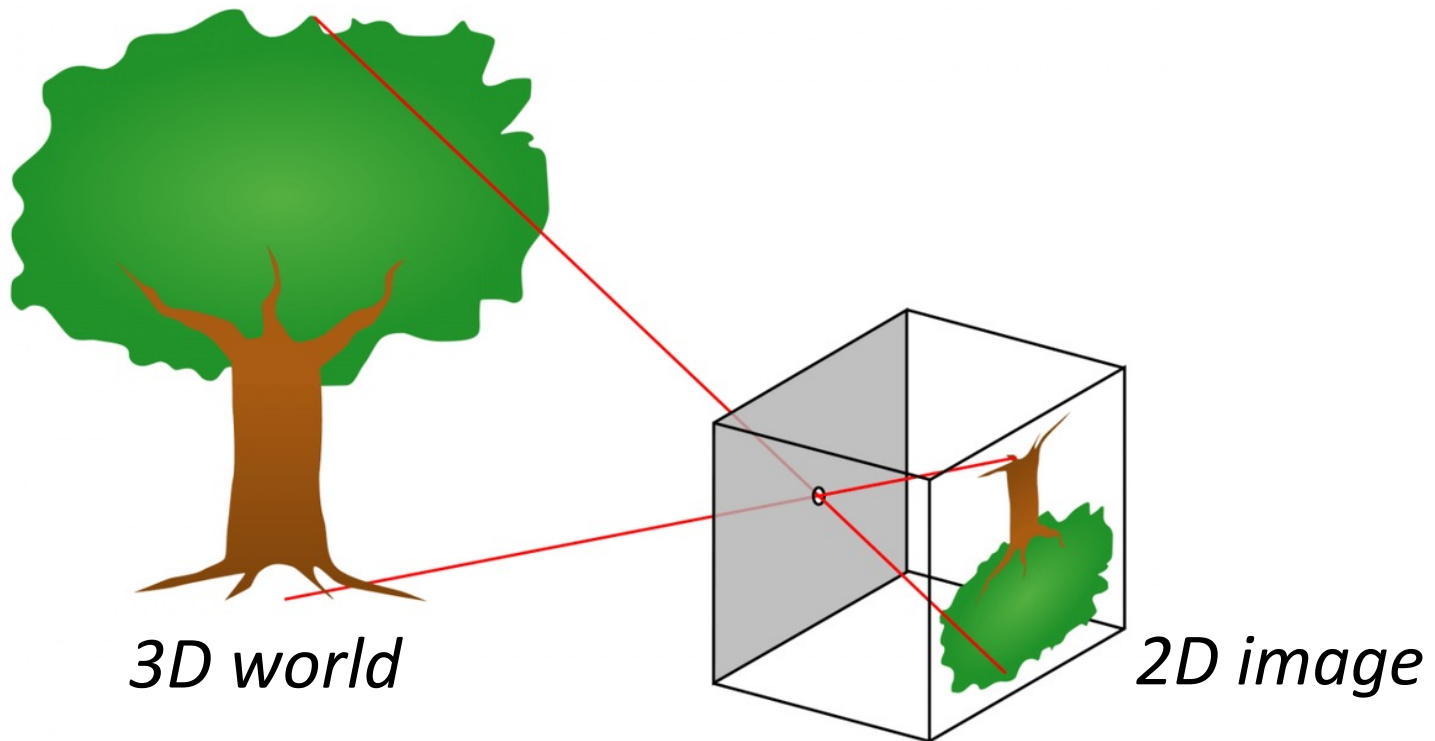


Today

- Camera and lens.
- **Geometric image formation.**
- Photometric image formation.

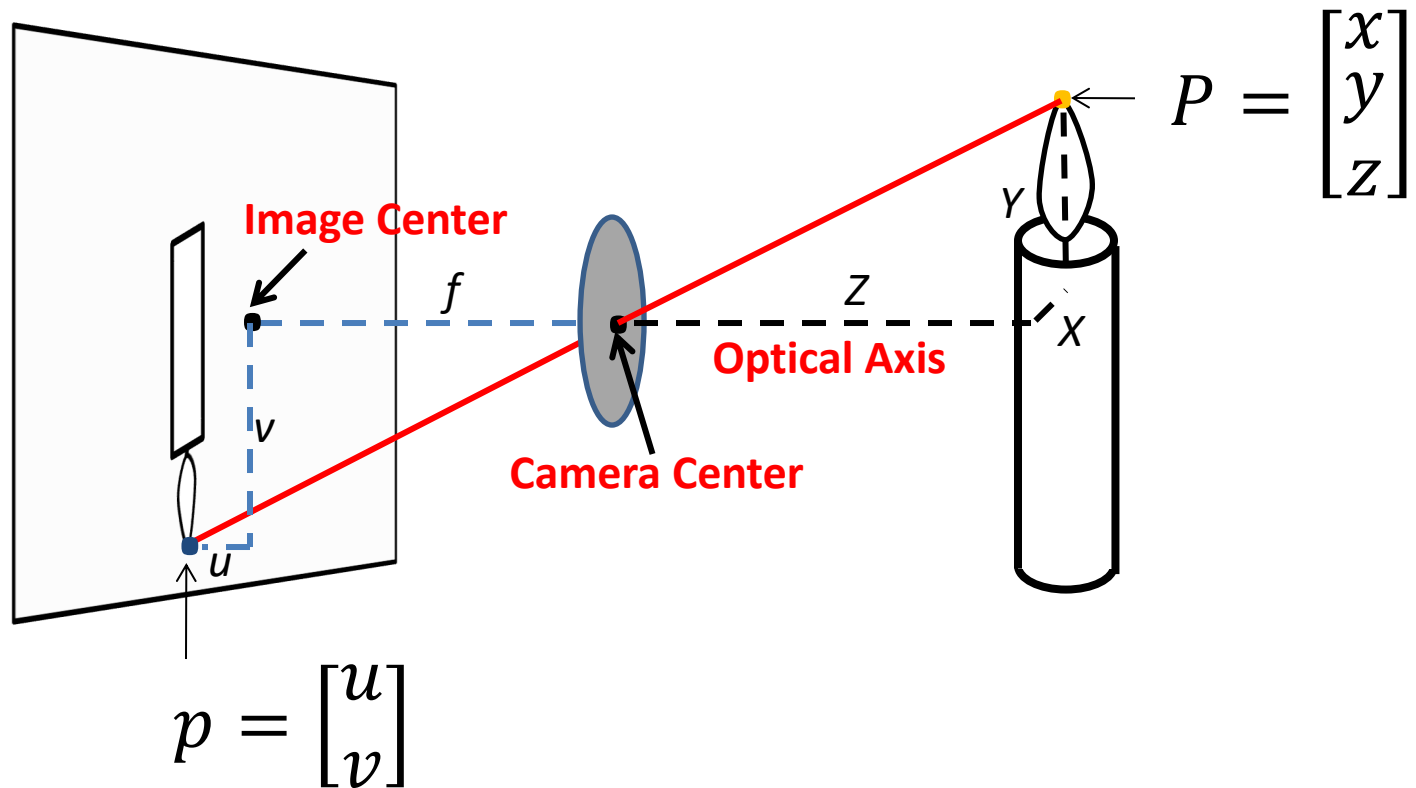
Geometric image formation

- Camera model describes the geometric relation between 3D world and 2D image



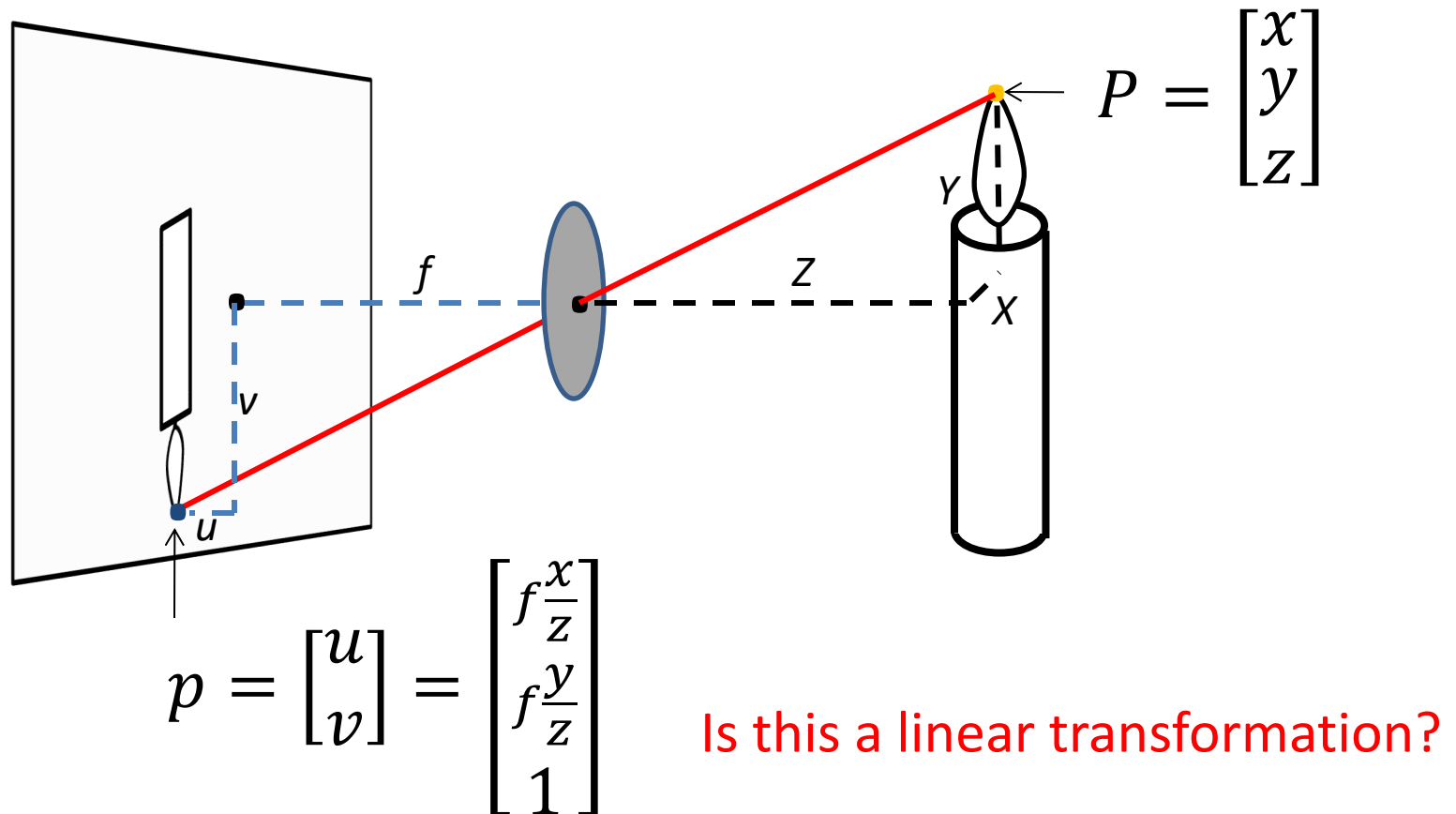
Pin-hole camera model: Perspective Projection

- 3D world coordinates $\rightarrow$ 2D image coordinates



Pin-hole camera model: Perspective Projection

- 3D world coordinates $\rightarrow$ 2D image coordinates



Homogeneous coordinates

Converting to homogeneous coordinates

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad (x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Converting from homogeneous to Cartesian

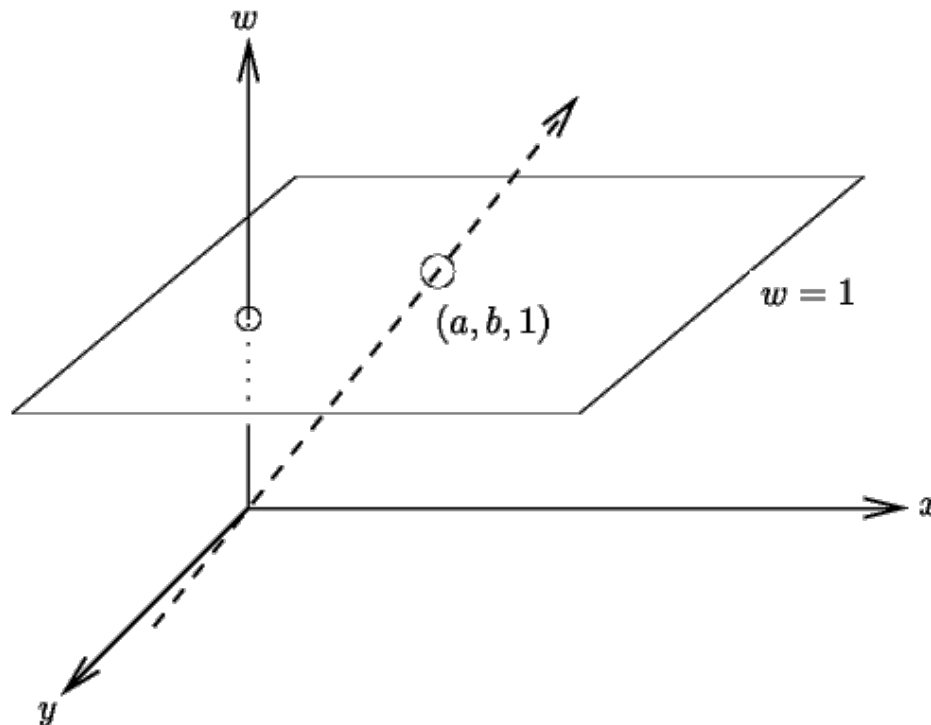
$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w) \qquad \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

Homogeneous coordinates are invariant to scaling

- Each point has an infinite set of homogeneous coordinates

Homogeneous coordinates

- Also called projective coordinates
 - The point in the 2D plane is projection of a ray in 3D space



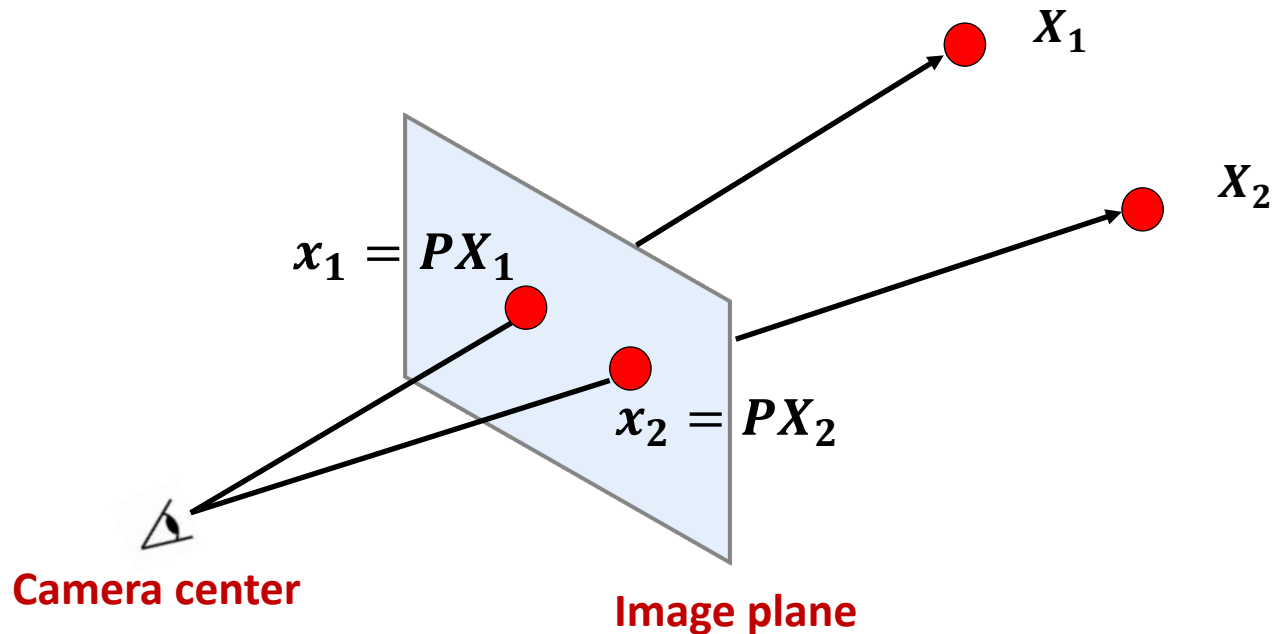
Perspective Projection in homogeneous coordinates

- Projection is a matrix multiplication using homogeneous coordinates

$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} fx \\ fy \\ z \end{bmatrix} \cong \begin{bmatrix} f\frac{x}{z} \\ f\frac{y}{z} \\ 1 \end{bmatrix}$$

Visualize Perspective Projection

- Put the image plane in front of the camera



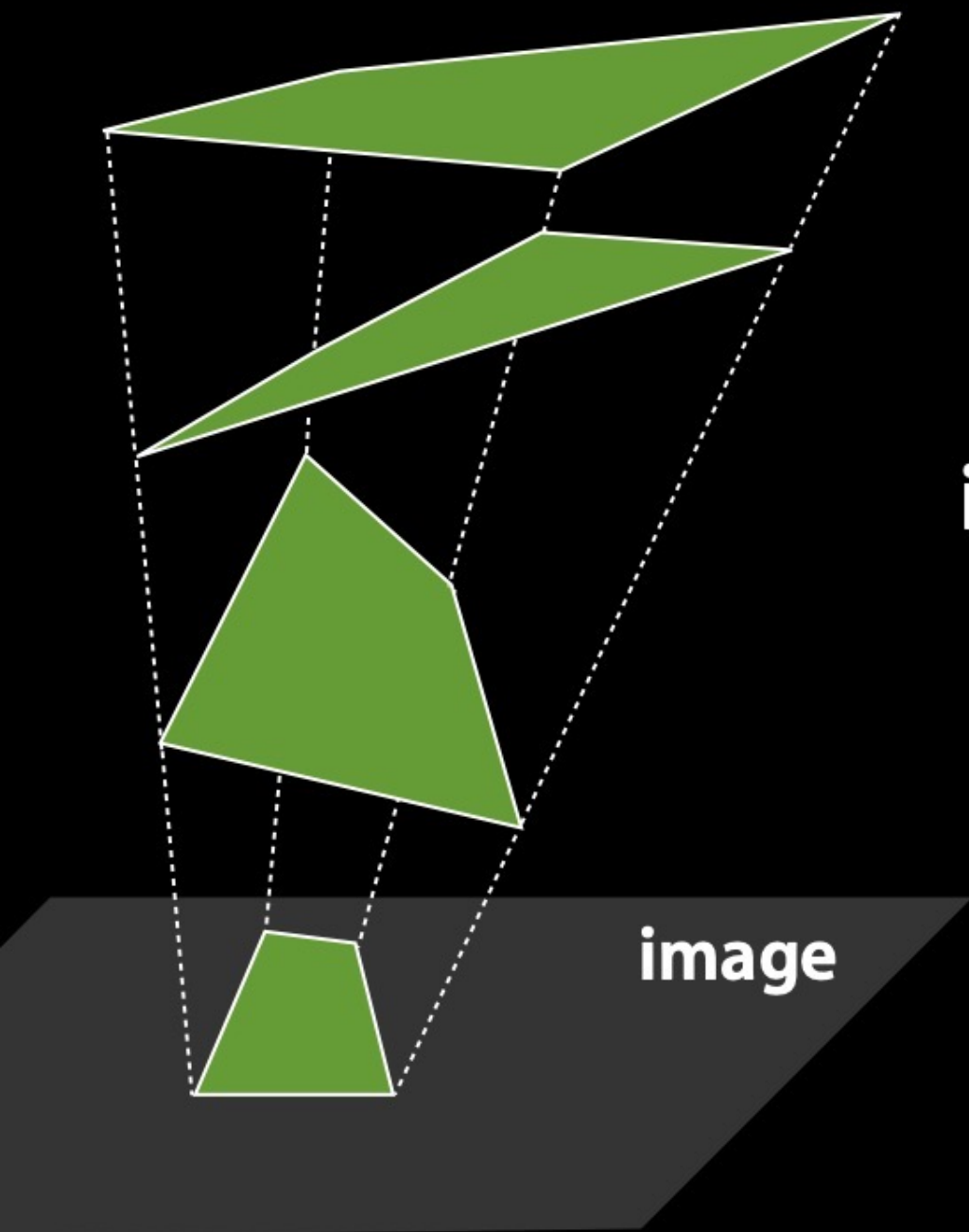
Projection can be tricky...



Projection can be tricky...



Making of 3D sidewalk art: <http://www.youtube.com/watch?v=3SNYtd0Ayt0>



**infinite number of
possible shapes**

Perspective Projection

What is preserved?

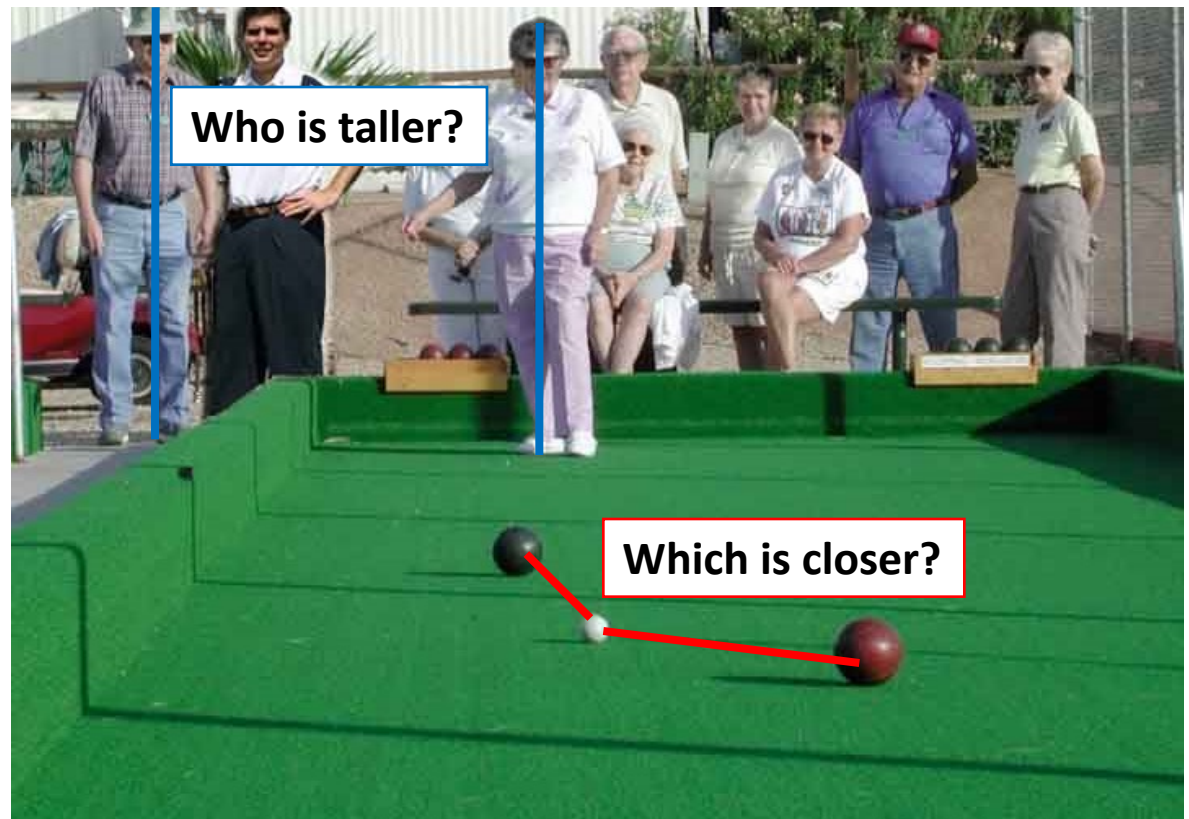
- Straight lines are still straight



Perspective Projection

What is lost?

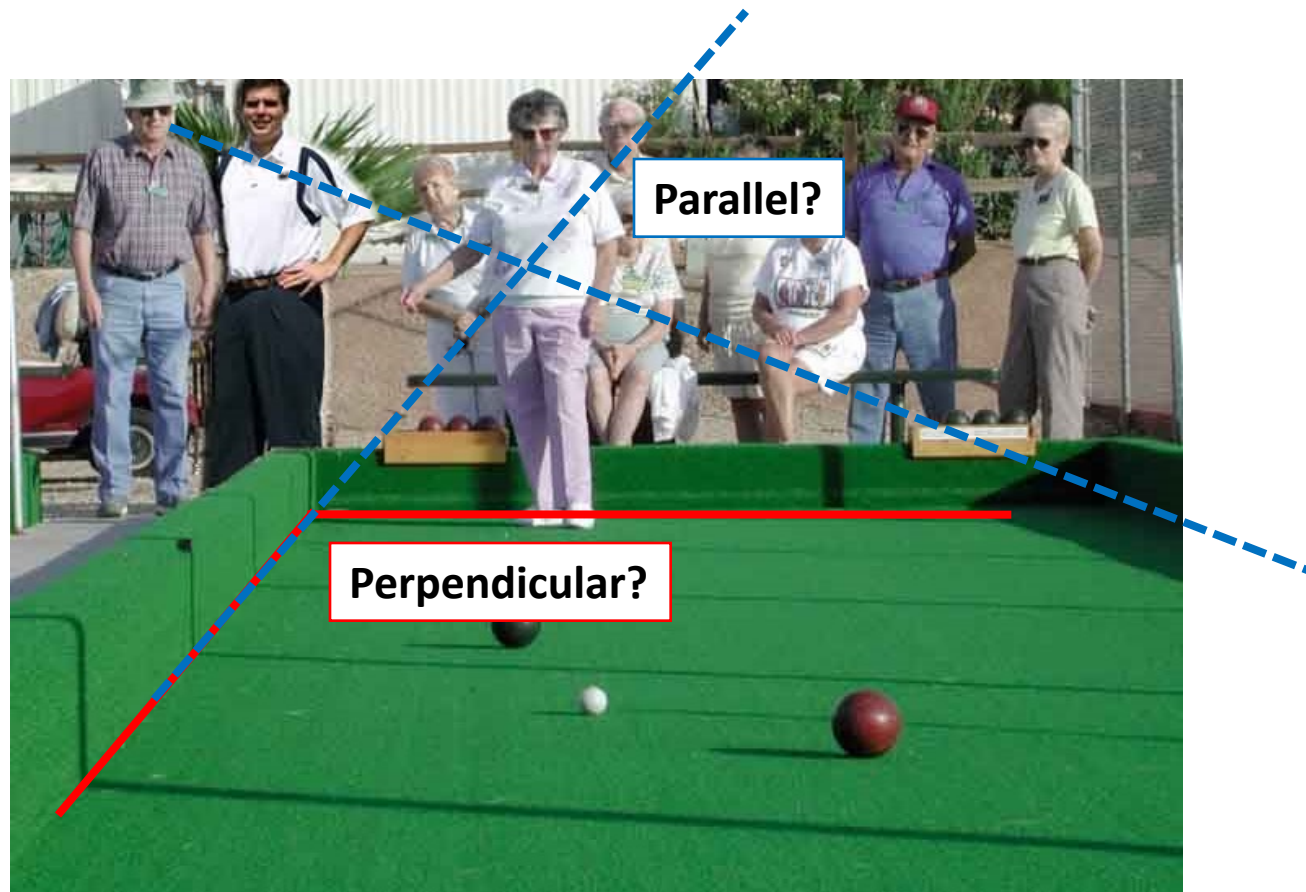
- Length



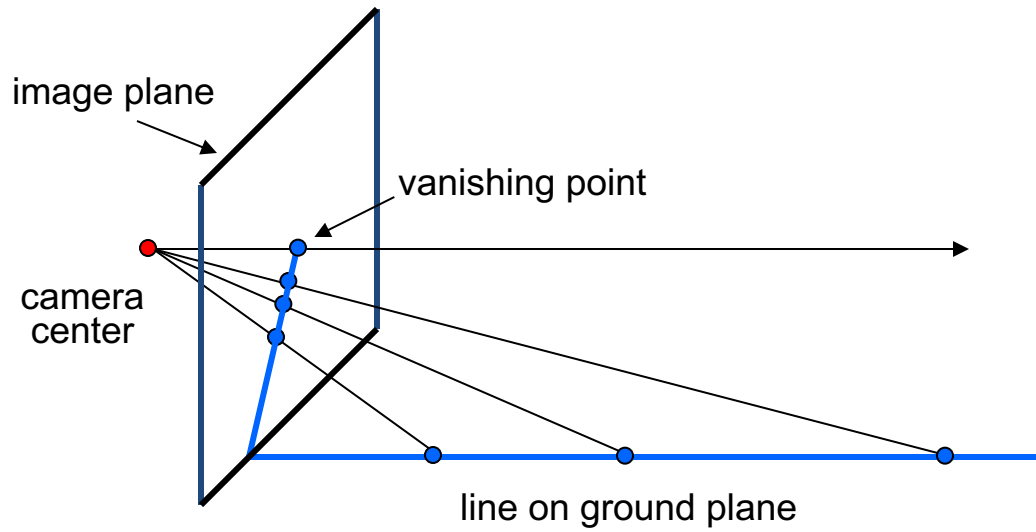
Perspective Projection

What is lost?

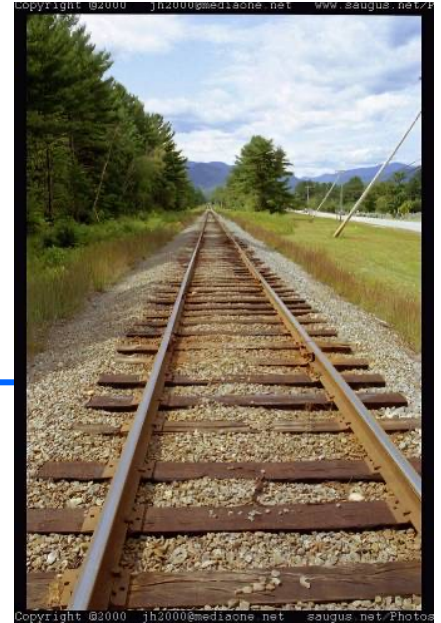
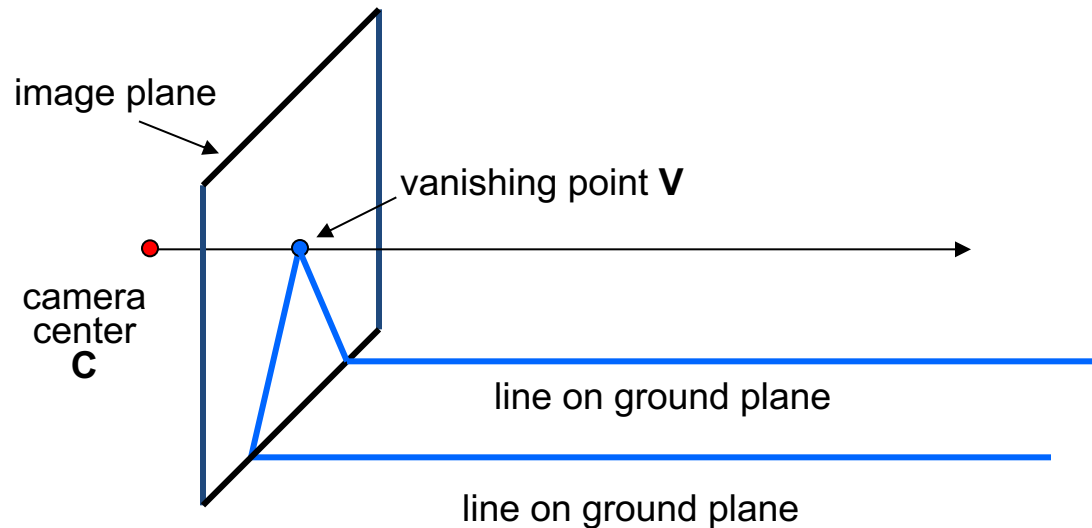
- Length
- Angles



Vanishing points

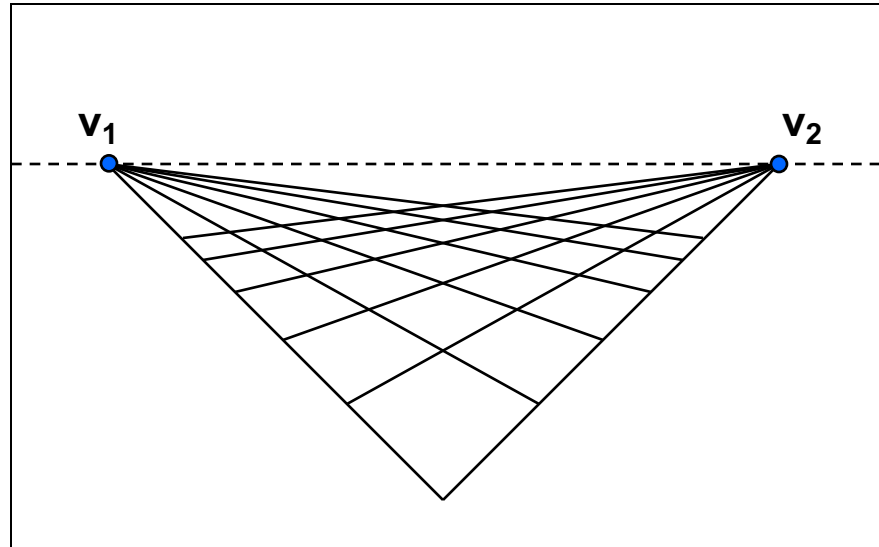


Vanishing points



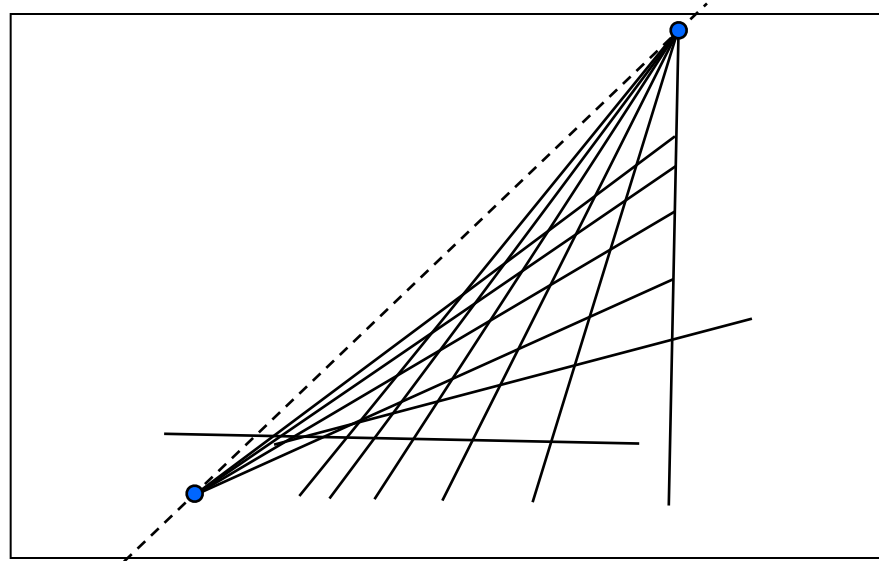
- Properties
 - Any two parallel lines have the same vanishing point v
 - The ray from C through v is parallel to the lines
 - v tells us the direction of the lines
 - v may be outside the image frame or at infinity

Vanishing lines



- Multiple Vanishing Points
 - Any set of parallel lines on the plane define a vanishing point
 - The union of all of these vanishing points is the vanishing line

Vanishing lines

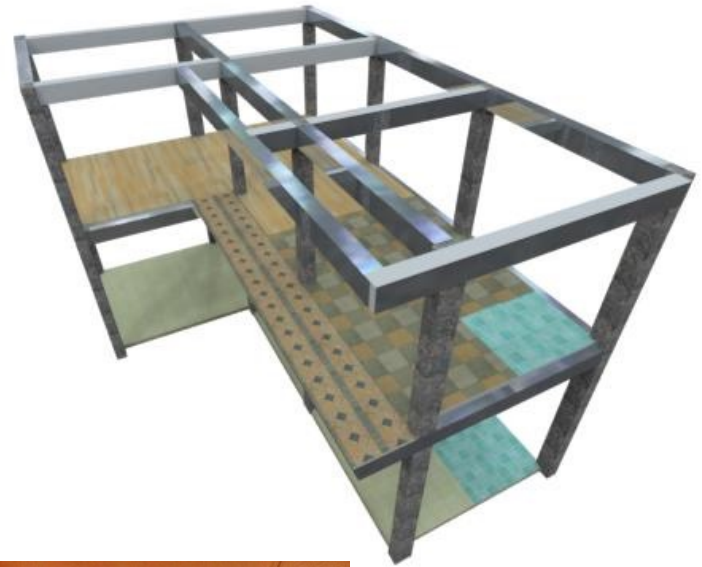


- Multiple Vanishing Points
 - Any set of parallel lines on the plane define a vanishing point
 - The union of all of these vanishing points is the vanishing line
 - Note that different planes define different vanishing lines
 - The direction of the vanishing line tells us the orientation of the plane

Horizon line



Perspective effect



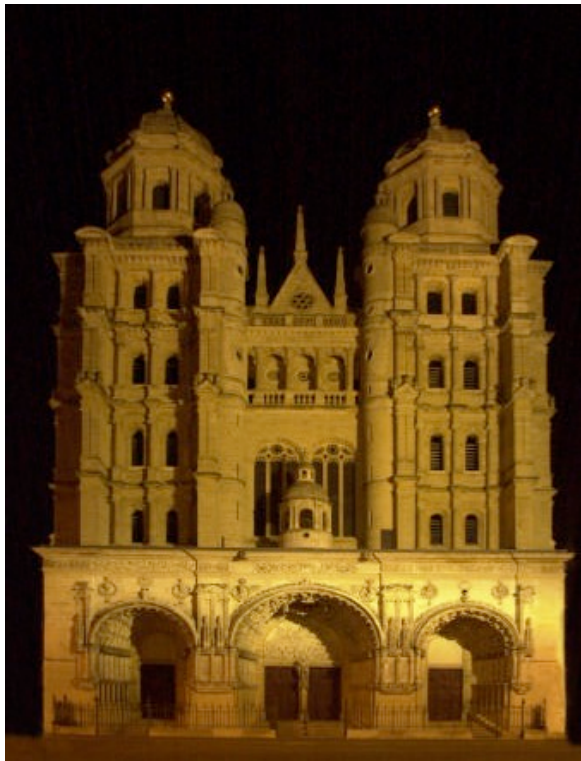
Perspective distortion

- Problem for architectural photography: converging verticals



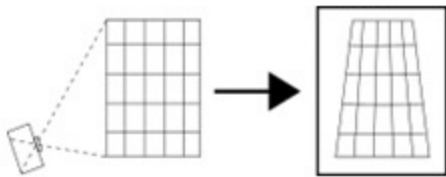
Perspective distortion

- Problem for architectural photography: converging verticals
- How to solve?

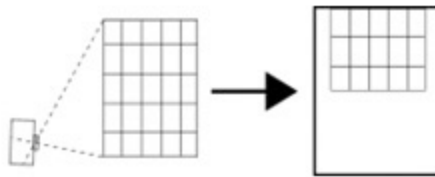


Perspective distortion

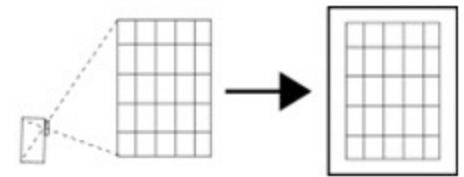
- Problem for architectural photography: converging verticals



Tilting the camera upwards results in converging verticals

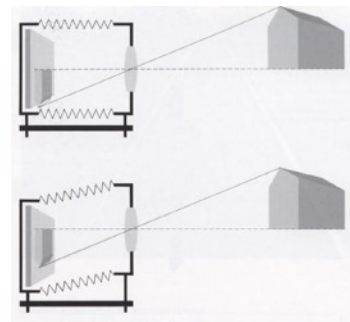
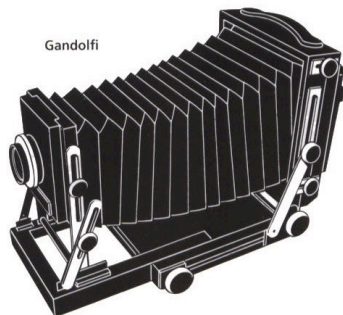


Keeping the camera level, with an ordinary lens, captures only the bottom portion of the building



Shifting the lens upwards results in a picture of the entire subject

- Solution: view camera (lens shifted w.r.t. film)



Perspective distortion

- What does a sphere project to?

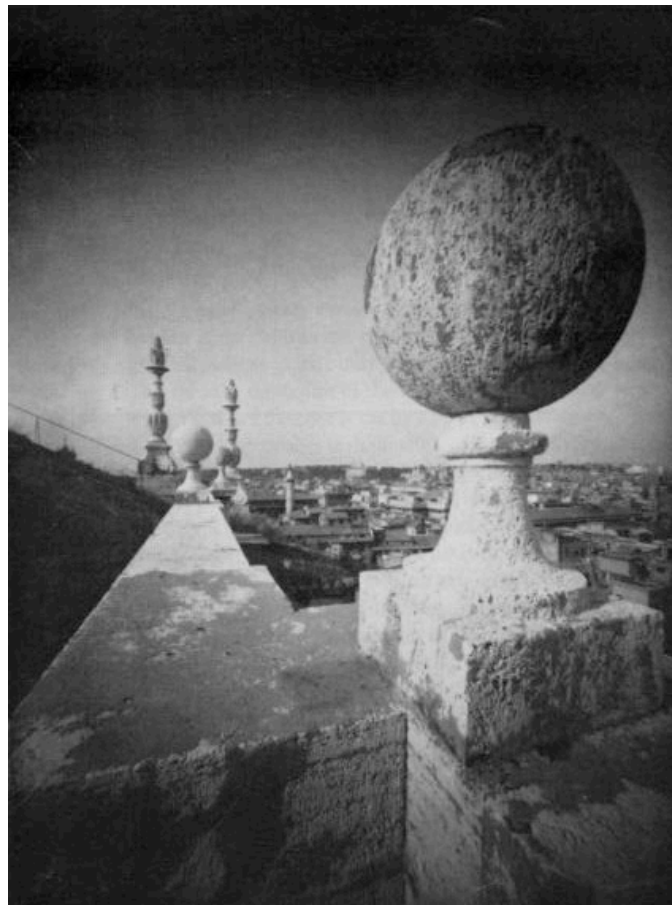
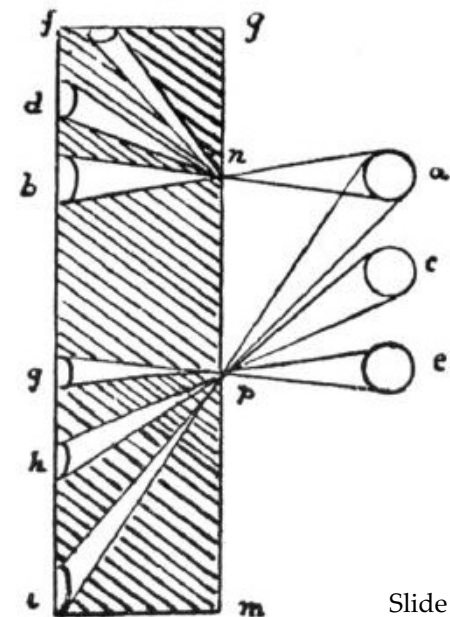
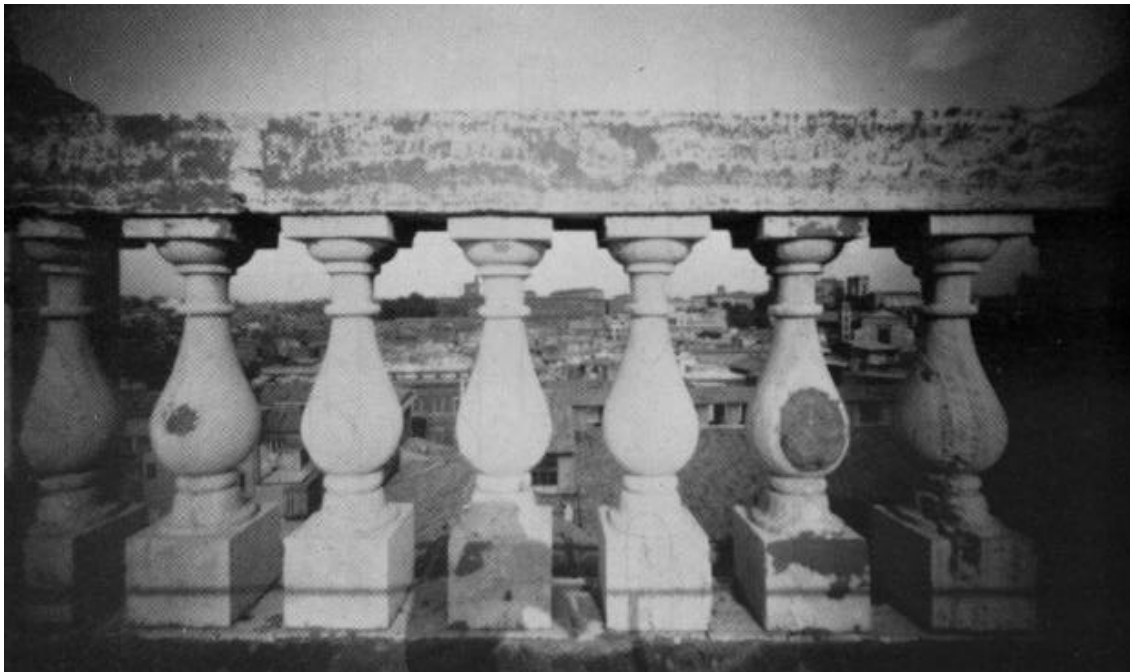


Image source: F. Durand

Perspective distortion

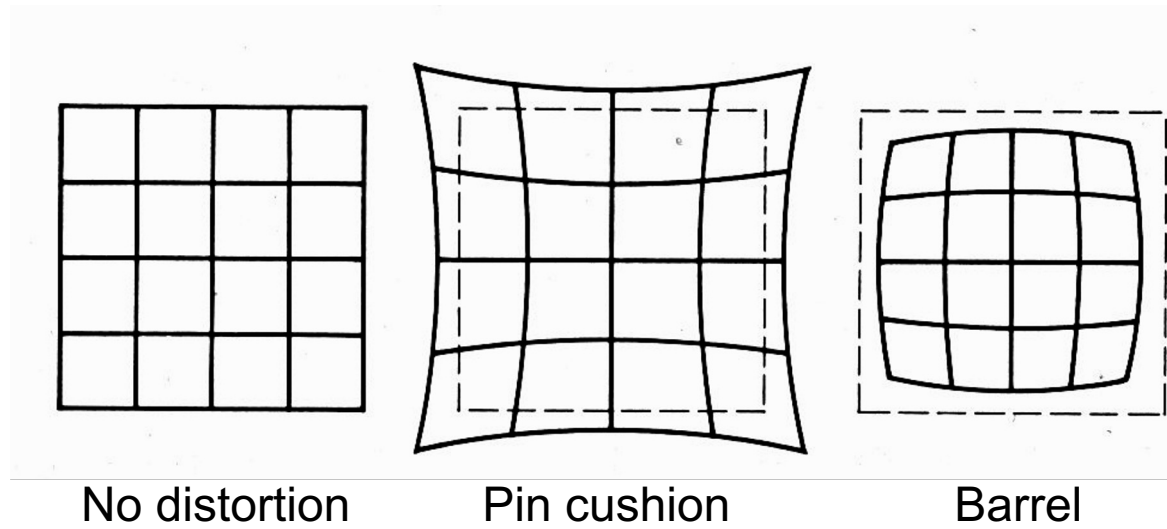
- The exterior columns appear bigger
- The distortion is not due to lens flaws
- Problem pointed out by Da Vinci



Perspective distortion: People

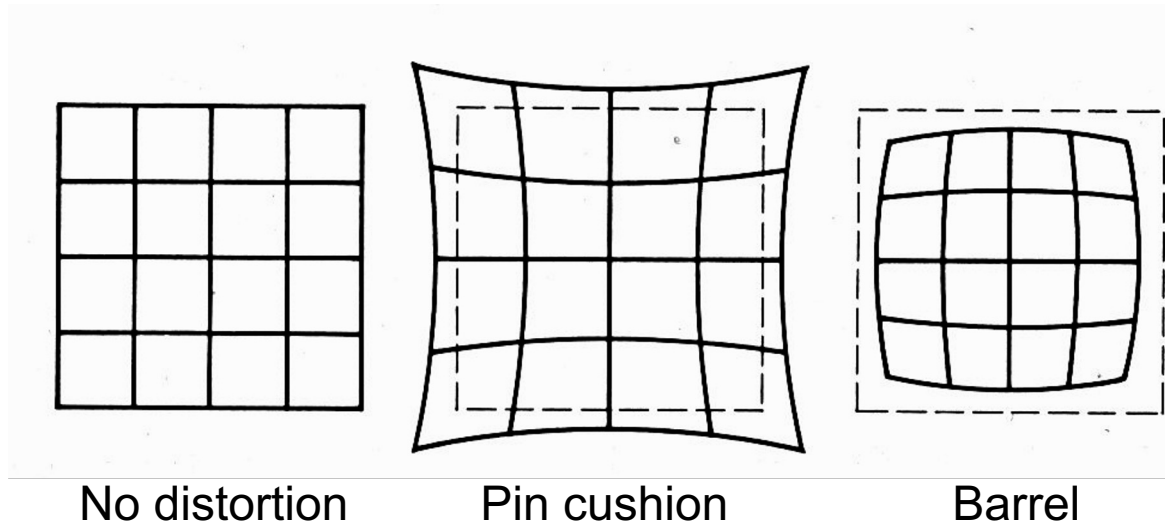


Distortion



- Radial distortion of the image
 - Caused by imperfect lenses
 - More noticeable for rays that pass through the edge of the lens

Distortion



$$r^2 = x_n'^2 + y_n'^2$$

$$x_d' = x_n'(1 + \kappa_1 r^2 + \kappa_2 r^4)$$

$$y_d' = y_n'(1 + \kappa_1 r^2 + \kappa_2 r^4)$$







Wide angle



Standard



Telephoto

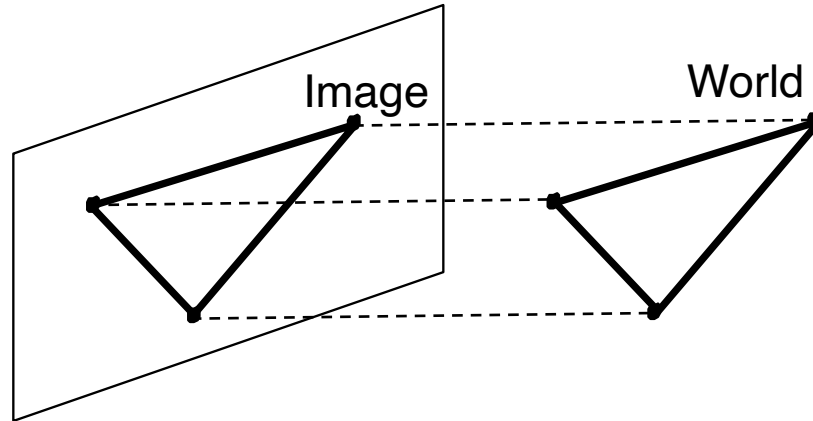
Correcting radial distortion



from [Helmut Dersch](#)

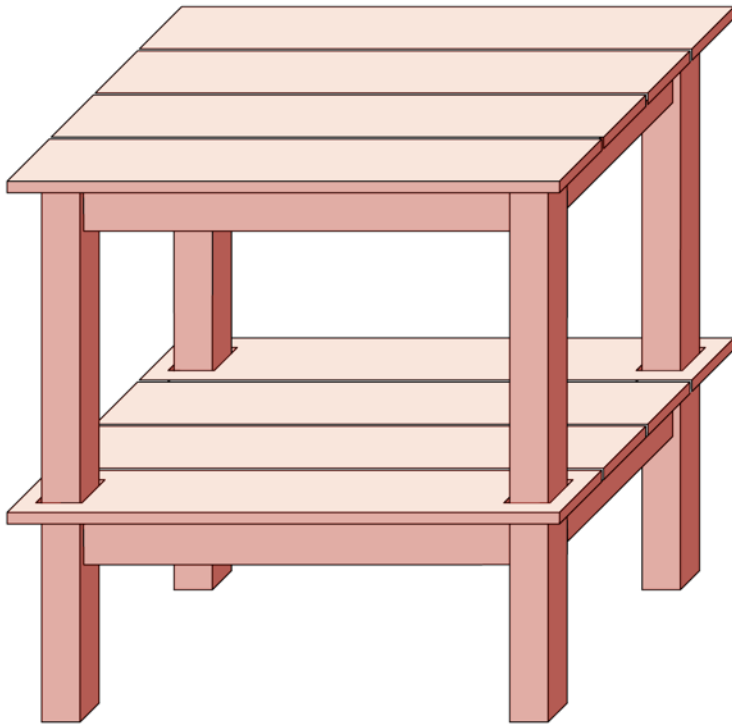
Orthographic projection

- Special case of perspective projection
 - Distance from the COP to the PP is infinite



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow (x, y)$$

Orthographic projection



Questions?

Today

- Camera and lens.
- Geometric image formation.
- **Photometric image formation.**

Photometric image formation

- Describes the relation between the physical properies of 3D world and the color of 2D image

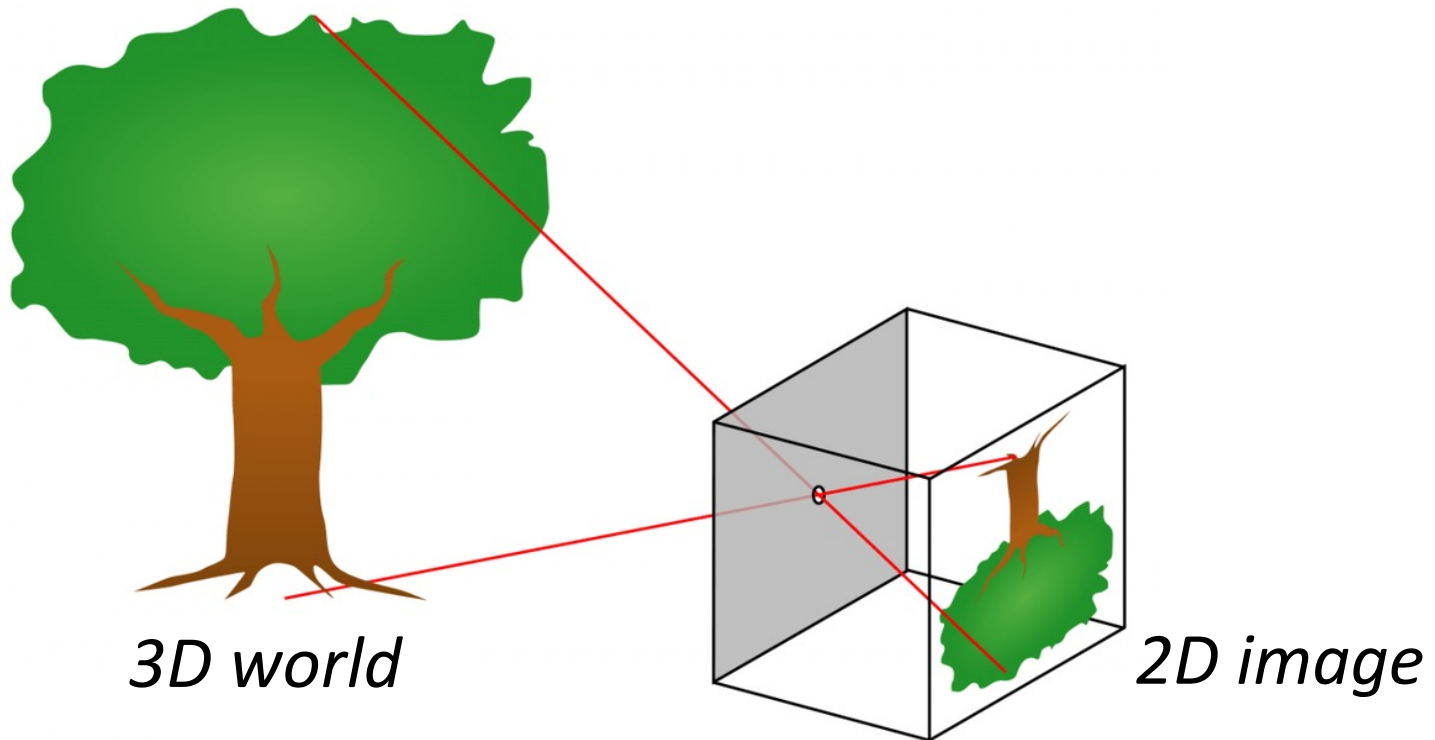
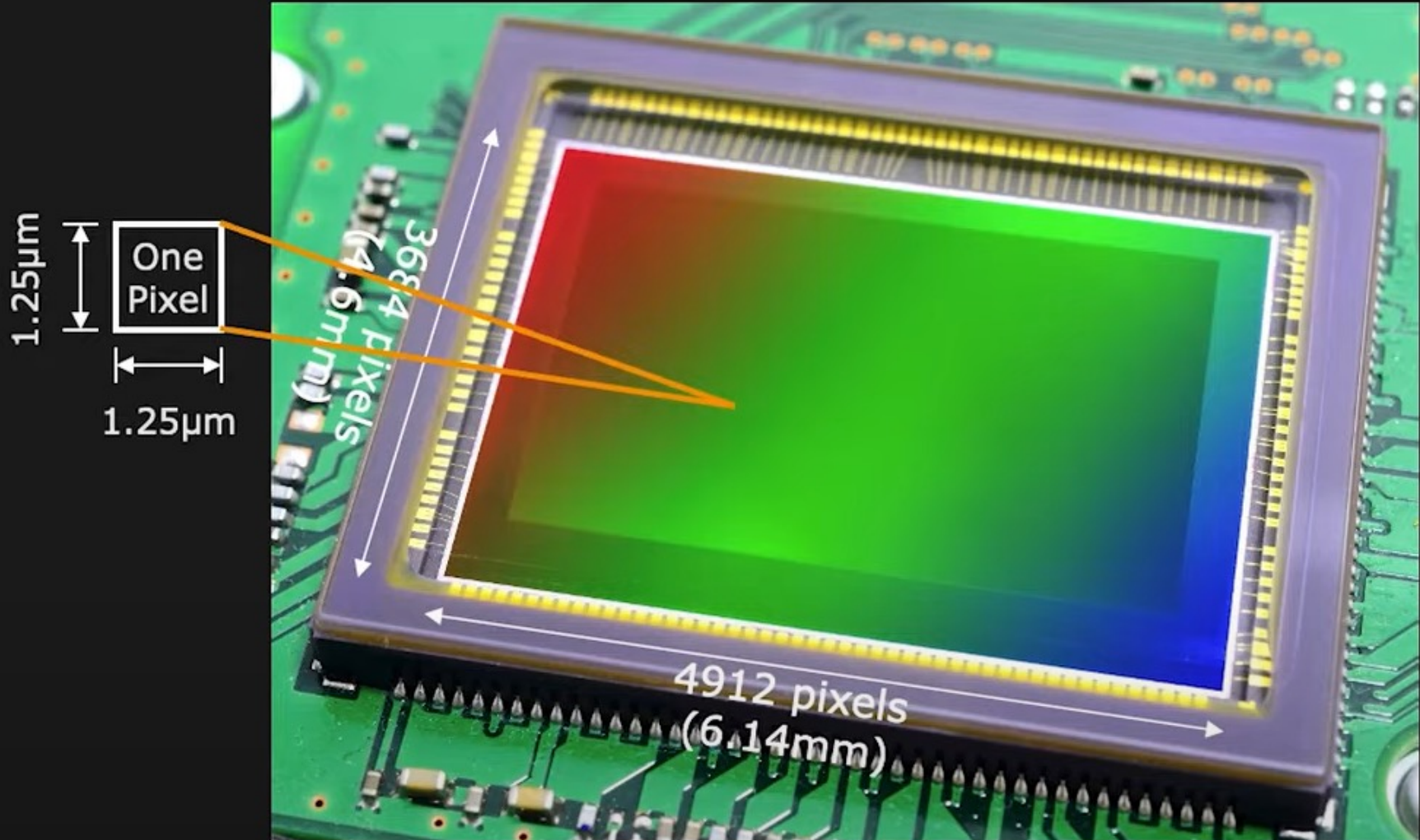
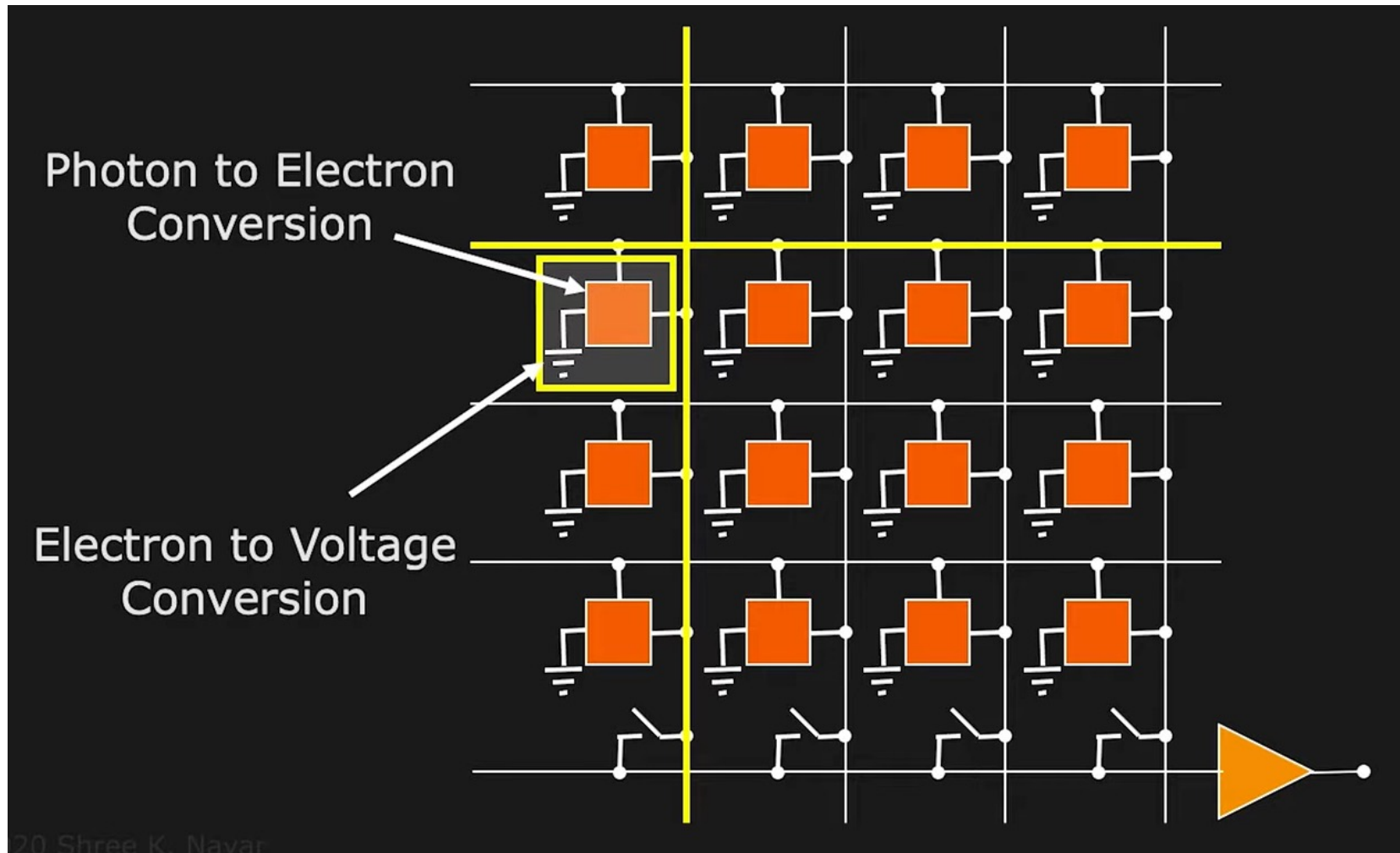


Image sensor

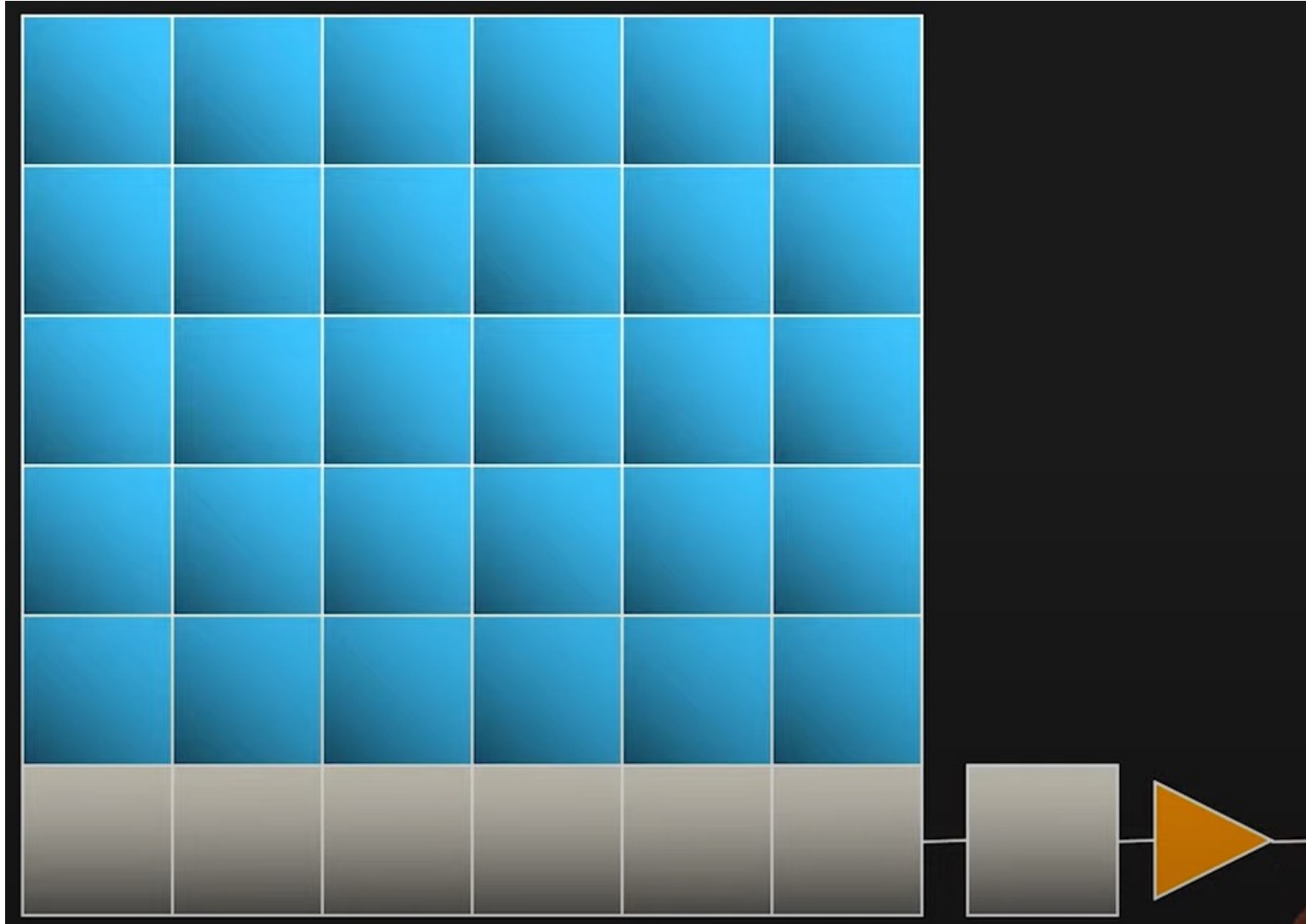


Credit: <https://fpcv.cs.columbia.edu/>

CMOS (Complimentary Metal-Oxide Semiconductor)



CCD (Charge Coupled Device)



Credit: <https://fpcv.cs.columbia.edu/>

Shutter

快门工作过程

当快门被触发后

1. 后幕帘开启
2. 前幕帘开启
--曝光开始
3. 后幕帘关闭
--曝光结束
4. 前幕帘关闭

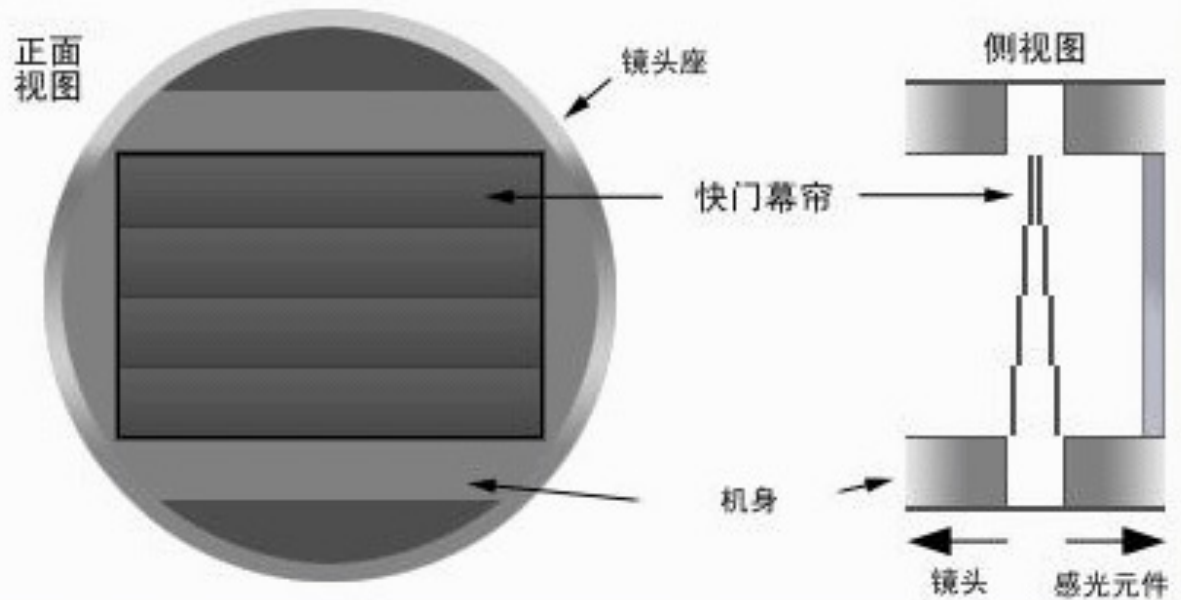


Fig 1.6.1: 快门工作过程

Shutter speed

- Shutter speed controls exposure time



Shutter speed

- Shutter speed controls exposure time
- The pixel value is equal to the integral of the light intensity within the exposure time



Shutter effect



Flowing water shot at different shutter speeds

Shutter effect



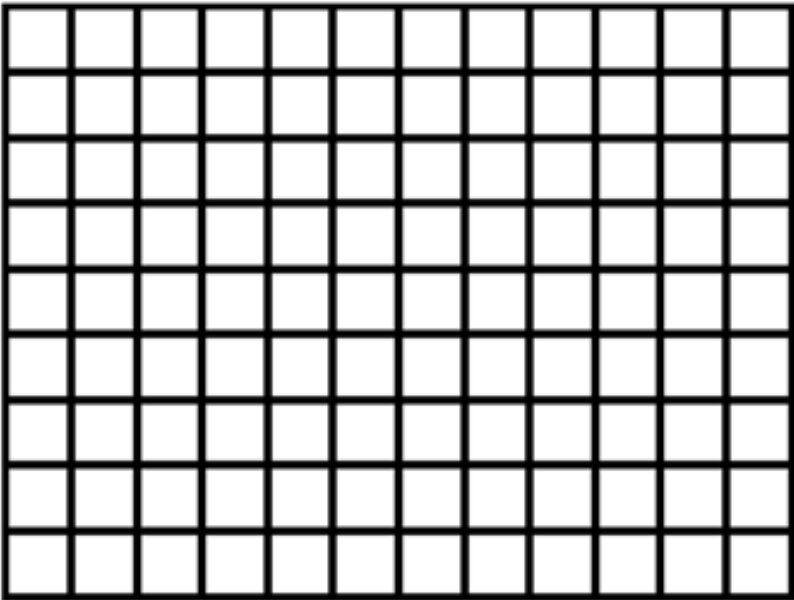
Light Paint

Rolling shutter effect

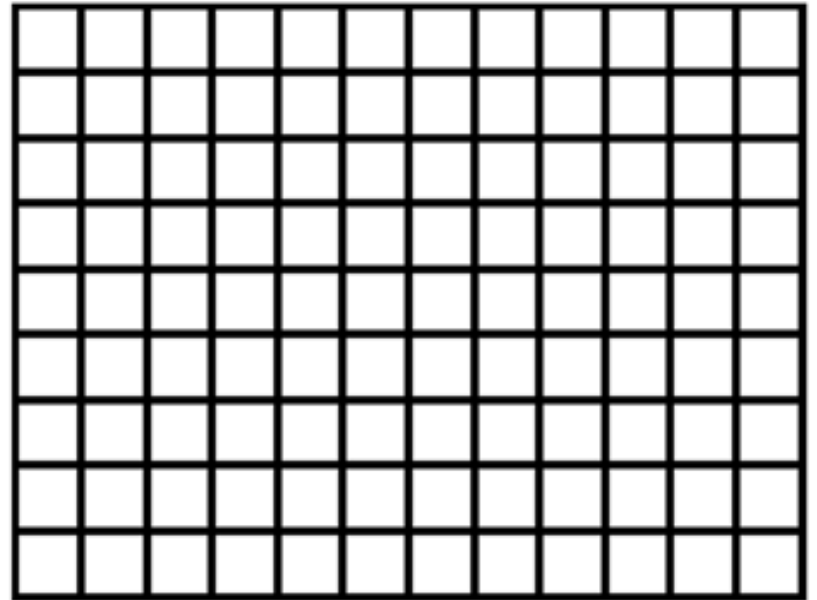


<https://www.lensrentals.com/blog/2021/07/understanding-the-difference-in-global-and-rolling-shutters/>

Rolling shutter effect



ROLLING SHUTTER



GLOBAL SHUTTER

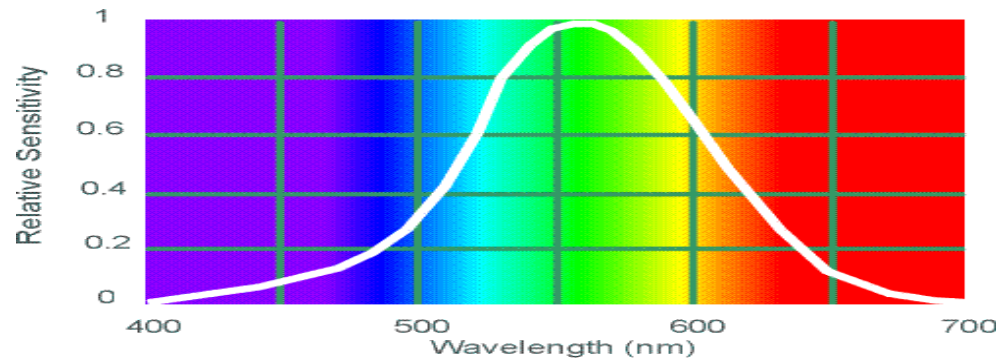
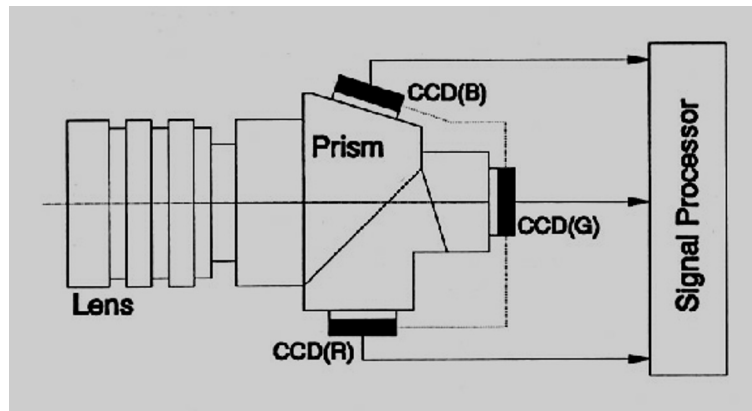
Rolling shutter effect



<https://www.lensrentals.com/blog/2021/07/understanding-the-difference-in-global-and-rolling-shutters/>

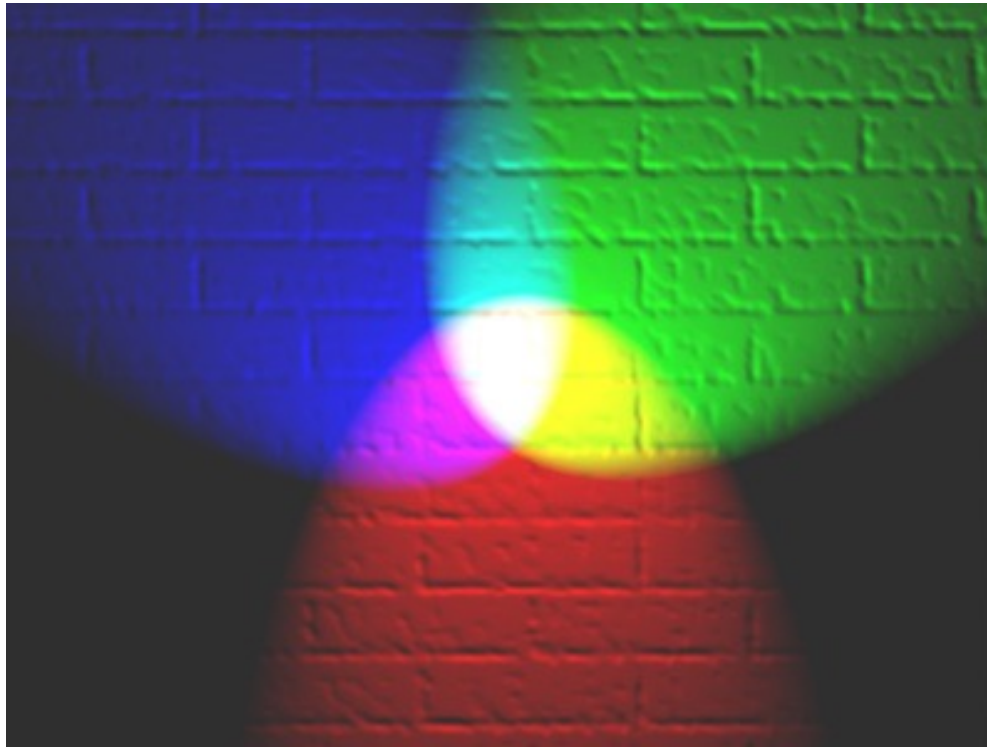
Color Sensing in Camera

- How to sense different colors?

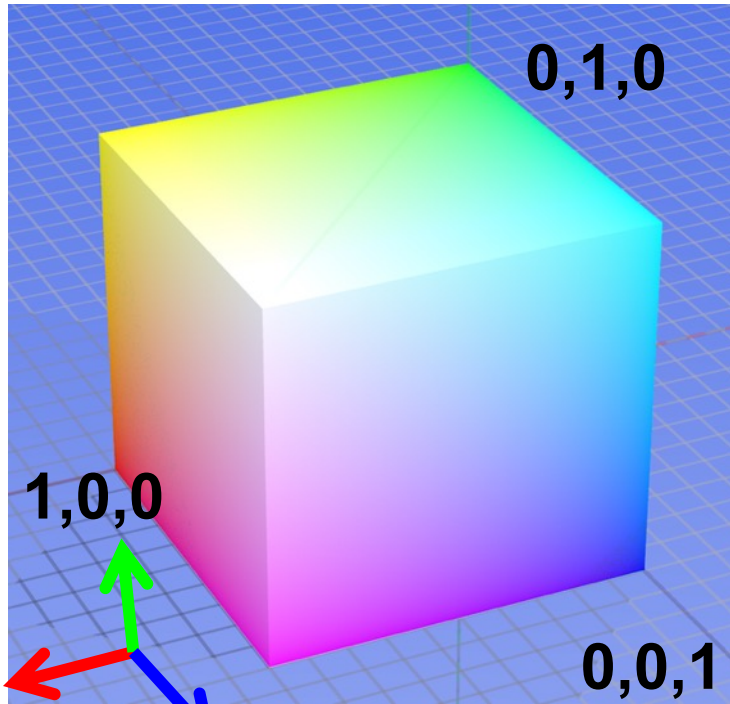


Color spaces

- How can we represent color?



Color spaces: RGB



RGB cube

- Easy for devices
- But not perceptual



R
(G=0,B=0)



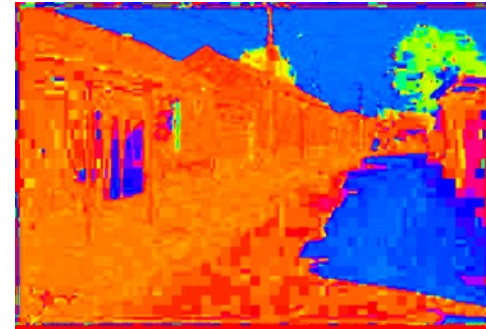
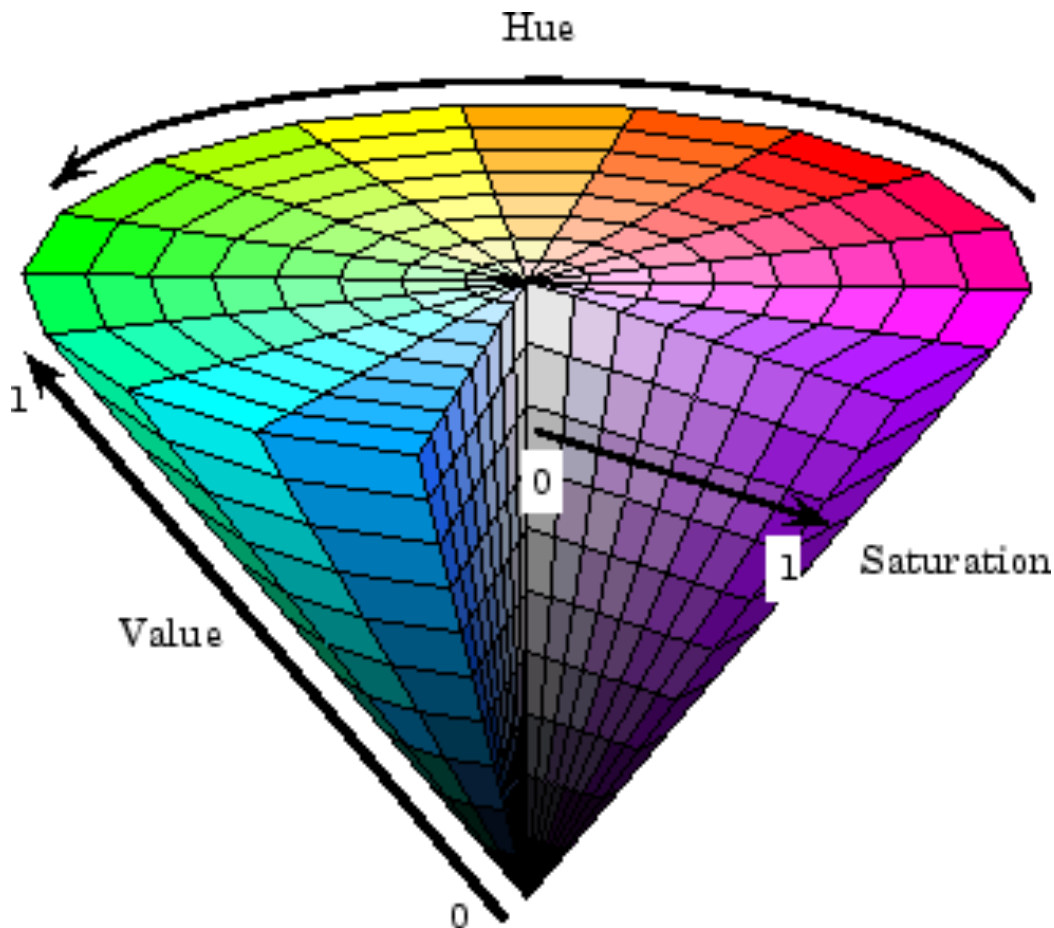
G
(R=0,B=0)



B
(R=0,G=0)

Color spaces: HSV

Intuitive color space



H
(S=1,V=1)



S
(H=1,V=1)



V
(H=1,S=0)

Color Image

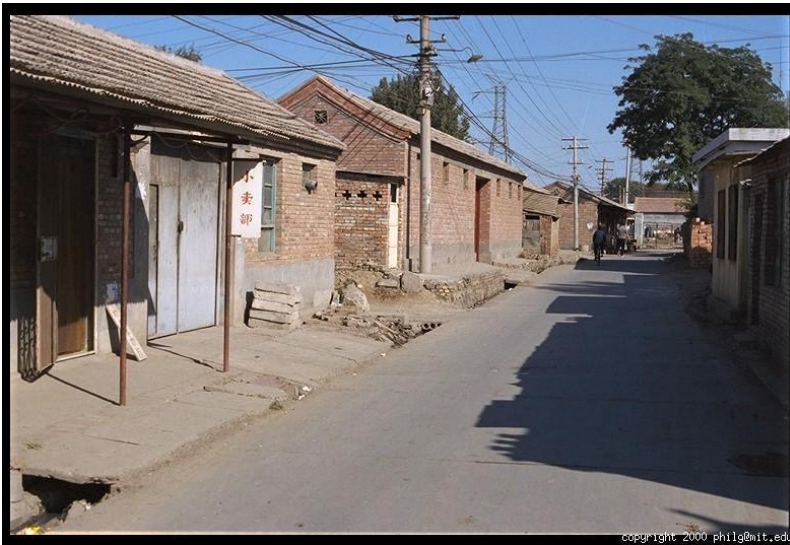
R



G



B



Images in Python

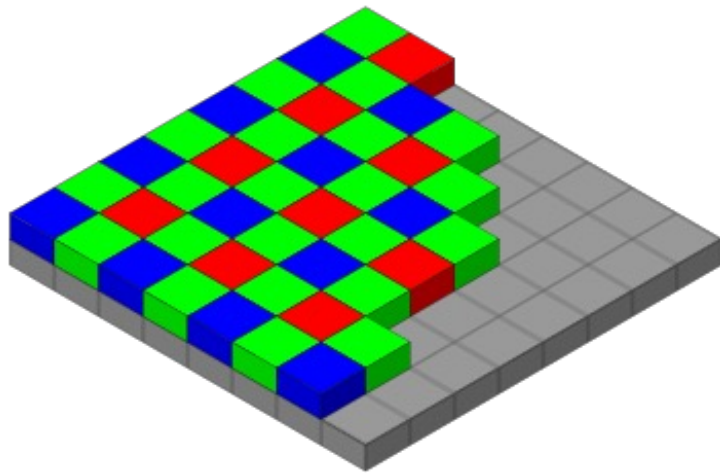
- Images represented as a matrix
- Suppose we have a $N \times M$ RGB image called “im”
 - $\text{im}(0,0,0)$ = top-left pixel value in R-channel
 - $\text{im}(y-1, x-1, b-1)$ = y pixels down, x pixels to right in the b^{th} channel
 - $\text{im}(N-1, M-1, 2)$ = bottom-right pixel in B-channel

The diagram illustrates the process of color conversion from RGB to CMYK. It shows three overlapping 10x10 grids of numerical values, representing the intensity of each color component (Red, Green, and Blue) at various points in the image. The grids are labeled R, G, and B. A blue arrow labeled 'column' points right, and another labeled 'row' points down, indicating the direction of processing.

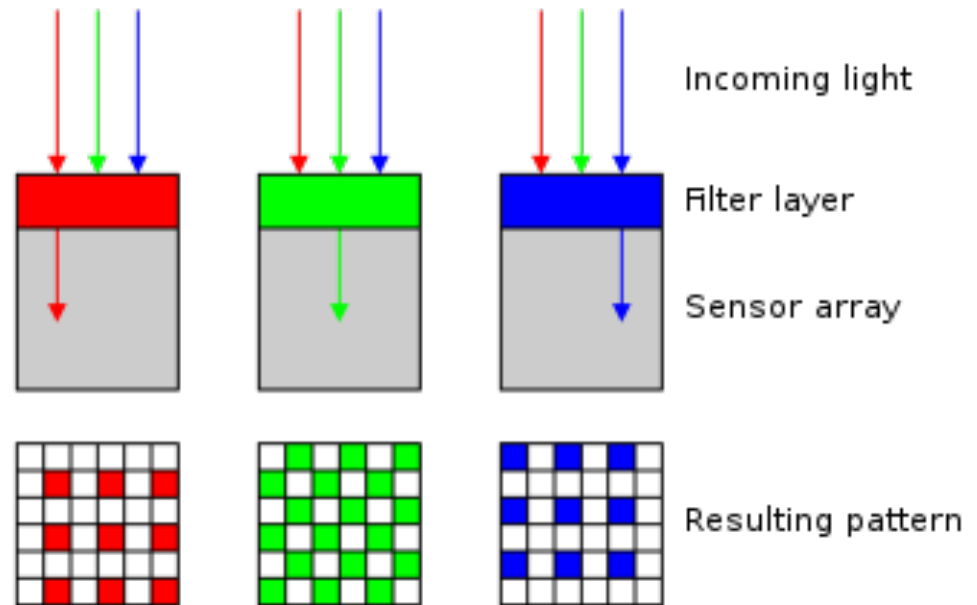
row \ column	1	2	3	4	5	6	7	8	9	10
1	0.92	0.93	0.94	0.97	0.62	0.37	0.85	0.97	0.93	0.92
2	0.95	0.89	0.82	0.89	0.56	0.31	0.75	0.92	0.81	0.95
3	0.89	0.72	0.51	0.55	0.51	0.42	0.57	0.41	0.49	0.91
4	0.96	0.95	0.88	0.94	0.56	0.46	0.91	0.87	0.90	0.97
5	0.71	0.81	0.81	0.87	0.57	0.37	0.80	0.88	0.89	0.79
6	0.49	0.62	0.60	0.58	0.50	0.60	0.58	0.50	0.61	0.45
7	0.86	0.84	0.74	0.58	0.51	0.39	0.73	0.92	0.91	0.49
8	0.96	0.67	0.54	0.85	0.48	0.37	0.88	0.90	0.94	0.82
9	0.69	0.49	0.56	0.66	0.43	0.42	0.77	0.73	0.71	0.90
10	0.79	0.73	0.90	0.67	0.33	0.61	0.69	0.79	0.73	0.93

Practical Color Sensing: Bayer filter

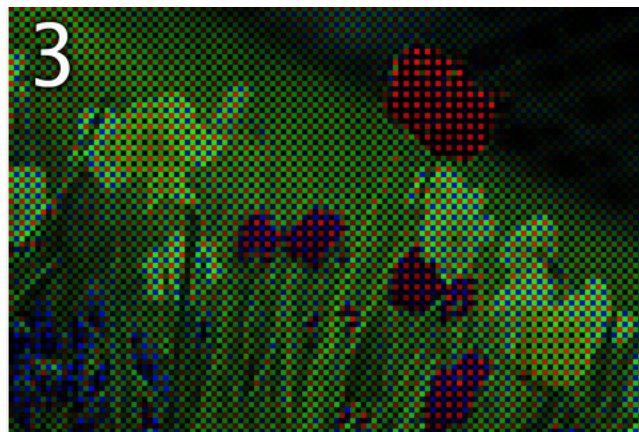
- Sensing RGB with a single chip with Bayer filter



Why more green?

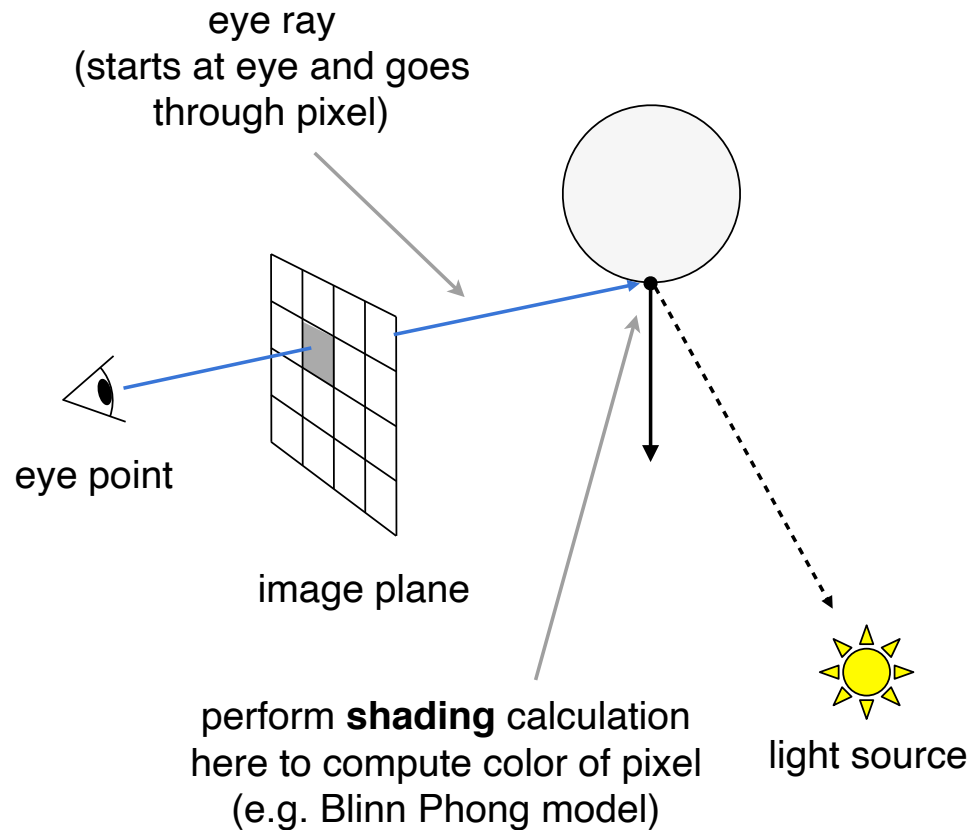


Practical Color Sensing: Bayer filter



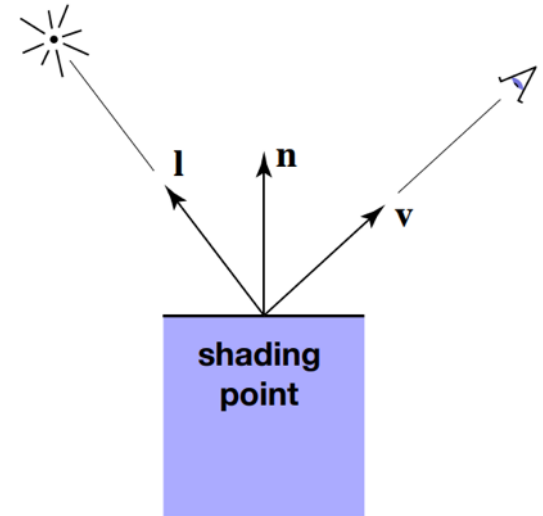
1. Original scene
2. Output of a 120×80-pixel sensor with a Bayer filter
3. Output color-coded with Bayer filter colors
4. Reconstructed image after interpolating missing color information

How to model the light reflected by an object?

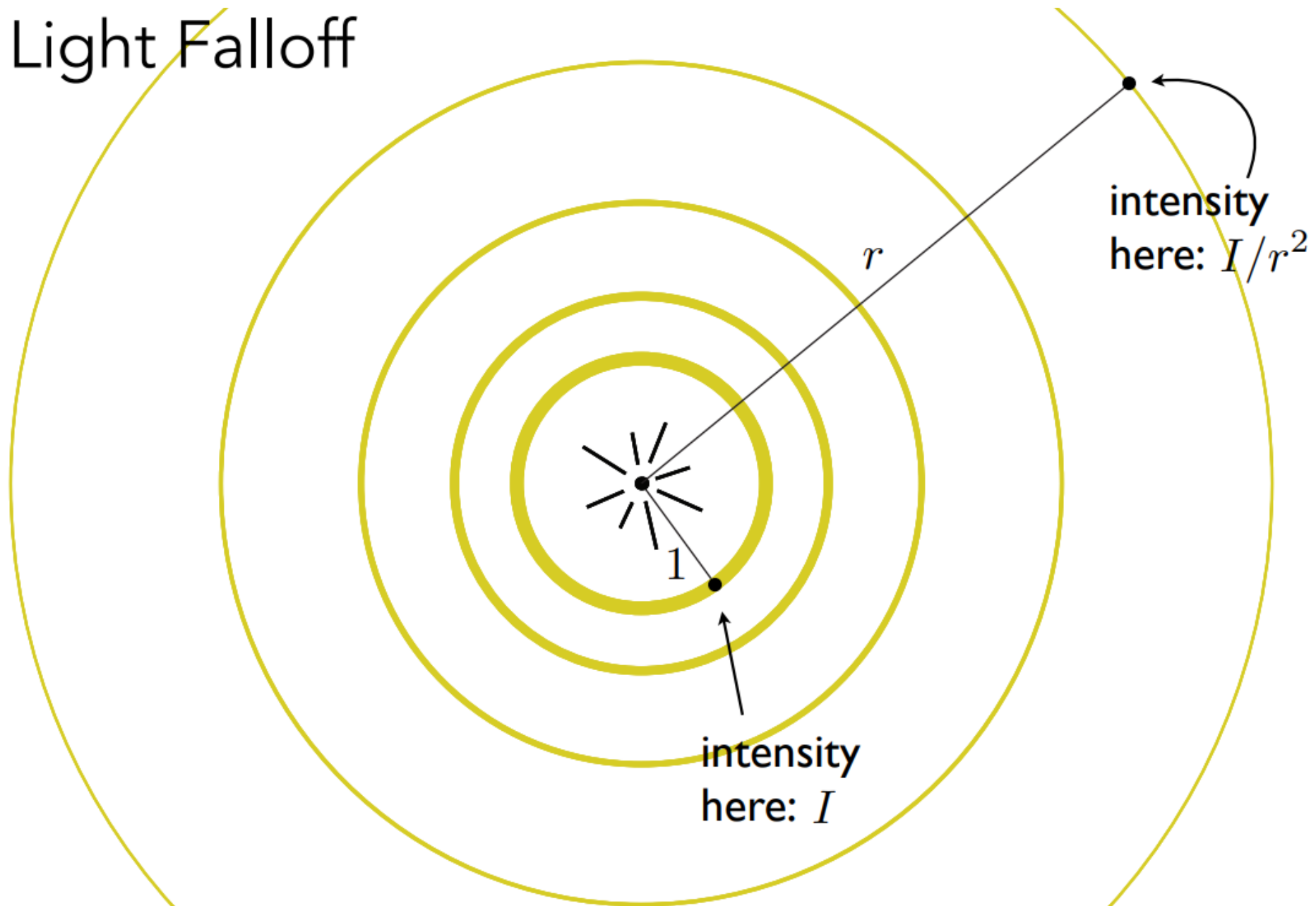


Shading

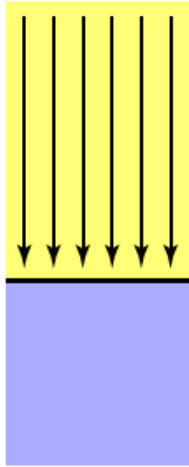
- Compute light reflected toward camera at a specific point.
- Inputs:
 - Viewer direction, v
 - Surface normal, n
 - Light direction, l (for each of many lights.)
 - Surface parameters



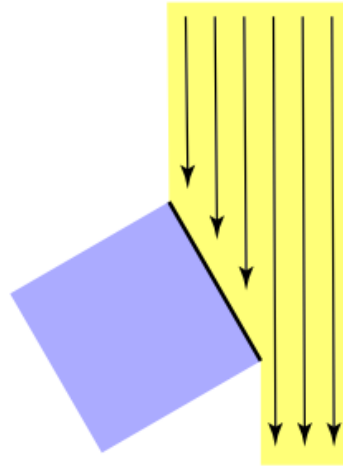
Light Falloff



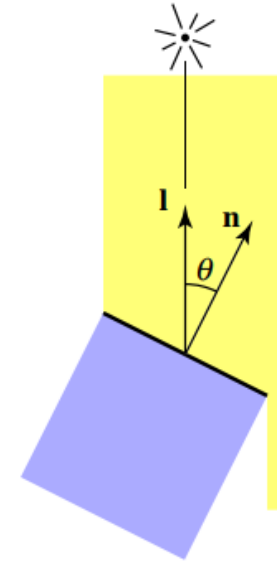
How much light (energy) is received?



Top face of cube
receives a certain
amount of light



Top face of
60° rotated cube
intercepts half the light



In general, light per unit
area is proportional to
 $\cos \theta = \mathbf{l} \cdot \mathbf{n}$

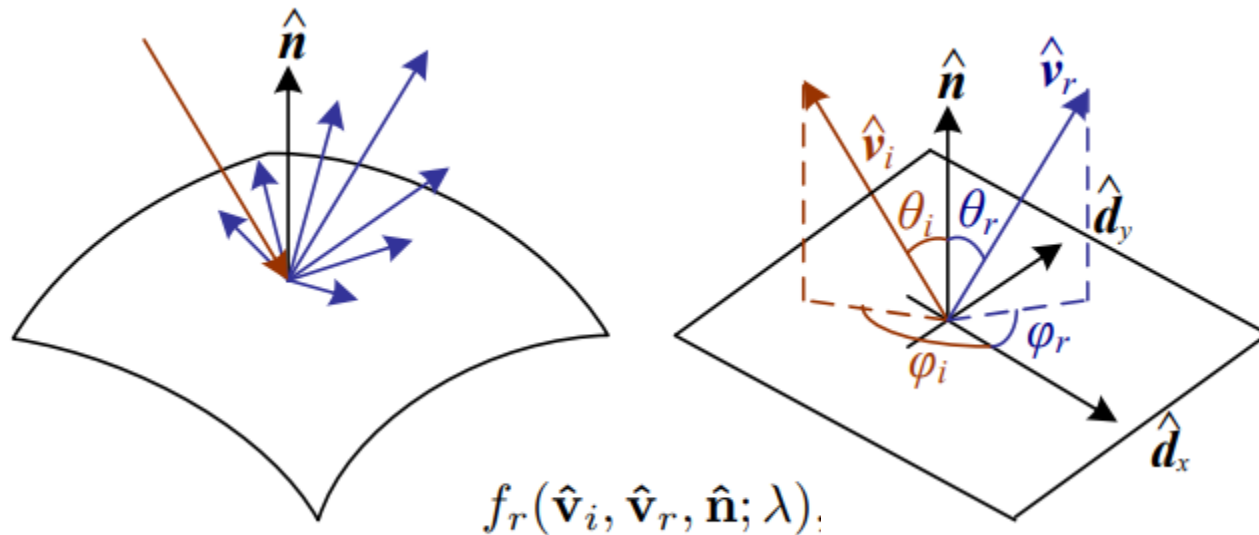
How light is reflected at a surface depends on material



Credit: <https://sites.cs.ucsb.edu/~lingqi/teaching/games101.html>

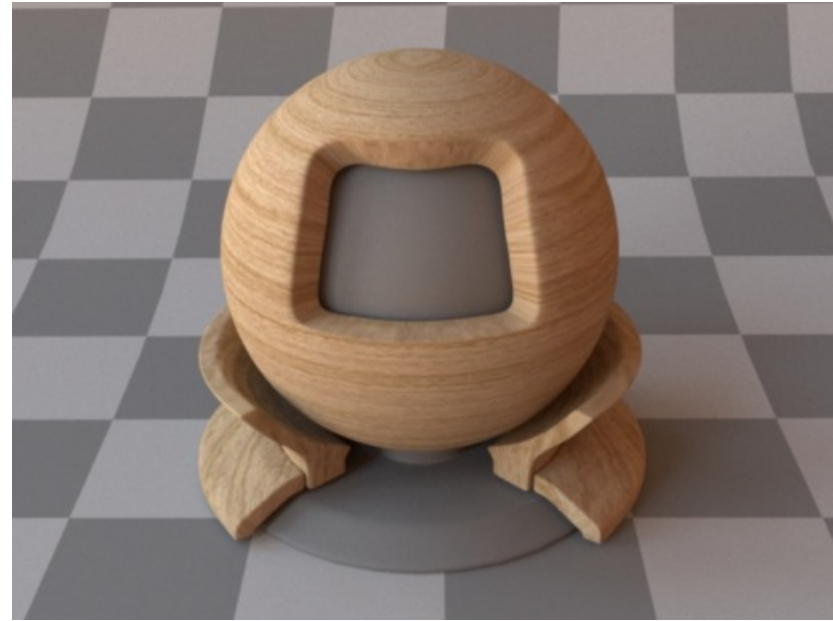
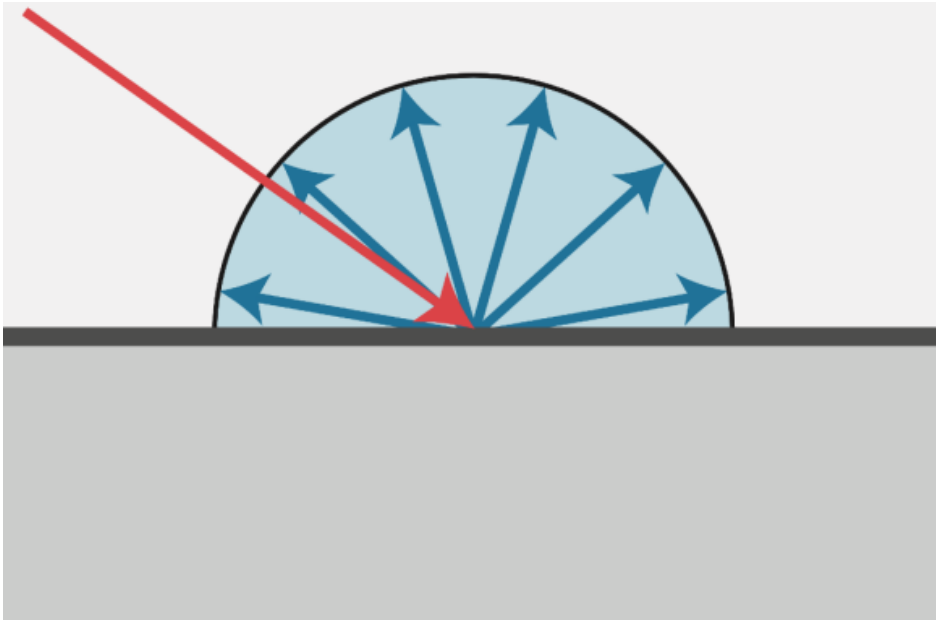
Material == BRDF

- Bidirectional reflectance distribution function: how light is reflected at a surface

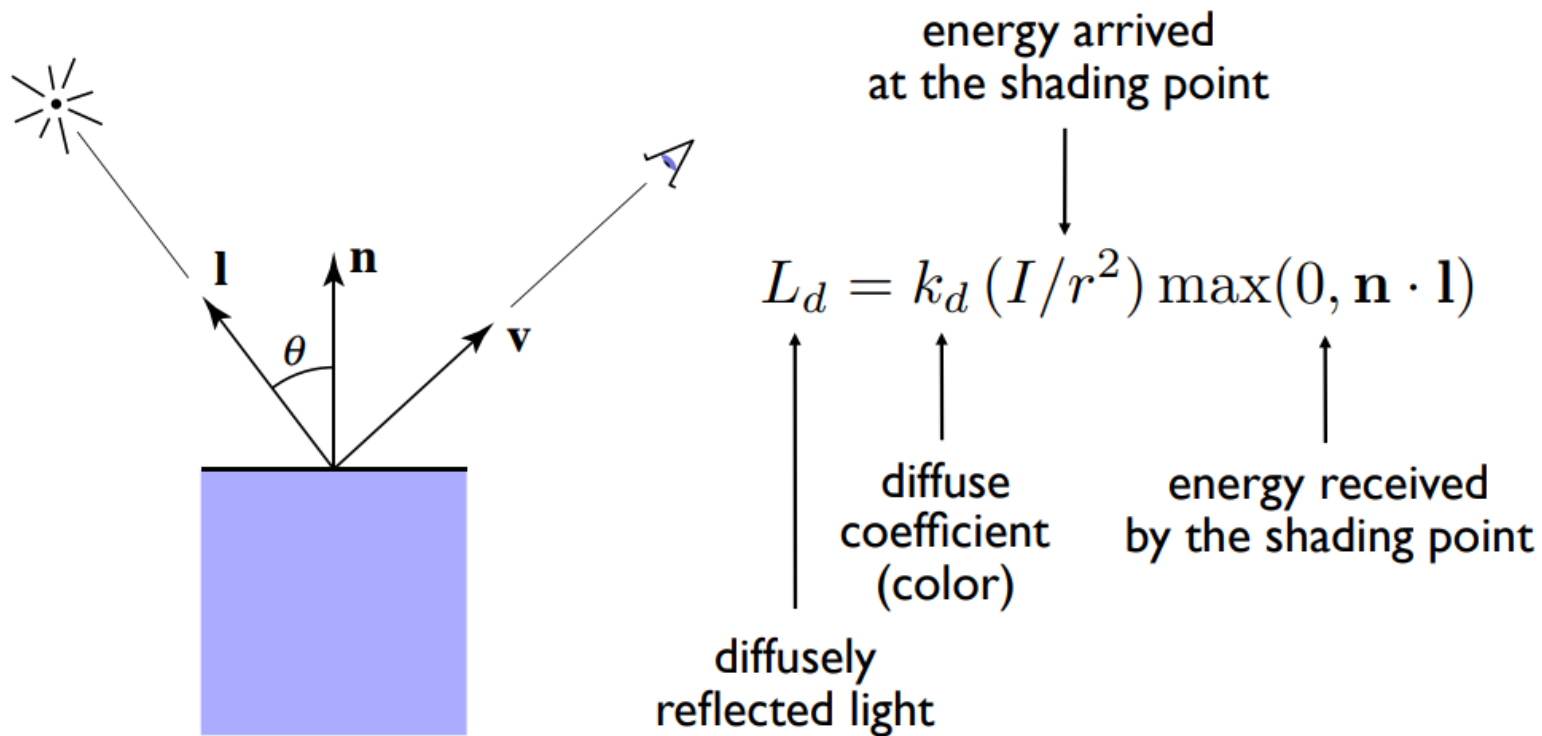


$$L_r(\hat{\mathbf{v}}_r; \lambda) = \int L_i(\hat{\mathbf{v}}_i; \lambda) f_r(\hat{\mathbf{v}}_i, \hat{\mathbf{v}}_r, \hat{\mathbf{n}}; \lambda) \cos^+ \theta_i d\hat{\mathbf{v}}_i$$

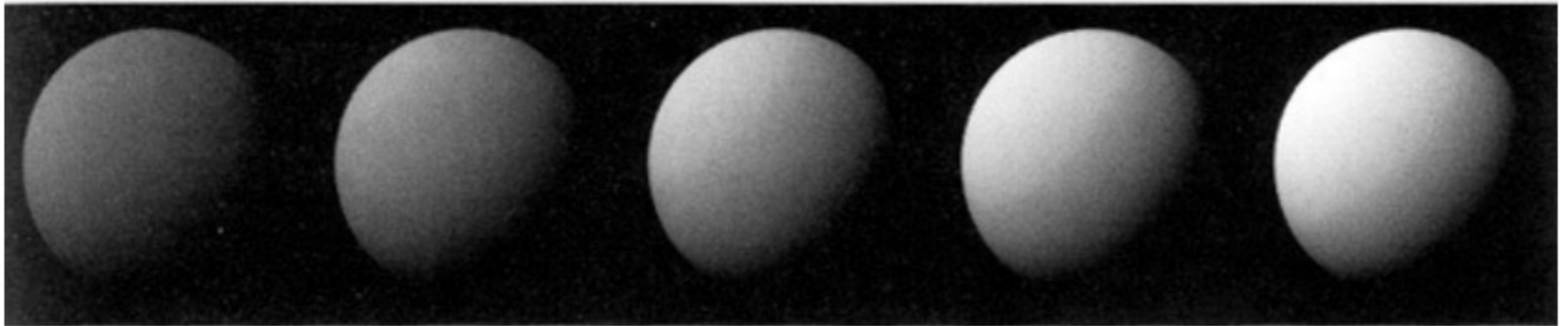
Diffuse (Lambertian) reflection



Diffuse (Lambertian) reflection



Diffuse (Lambertian) reflection

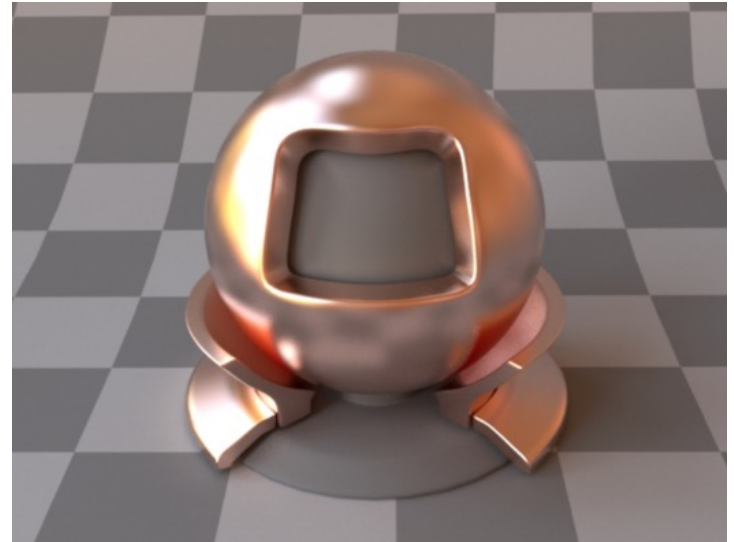
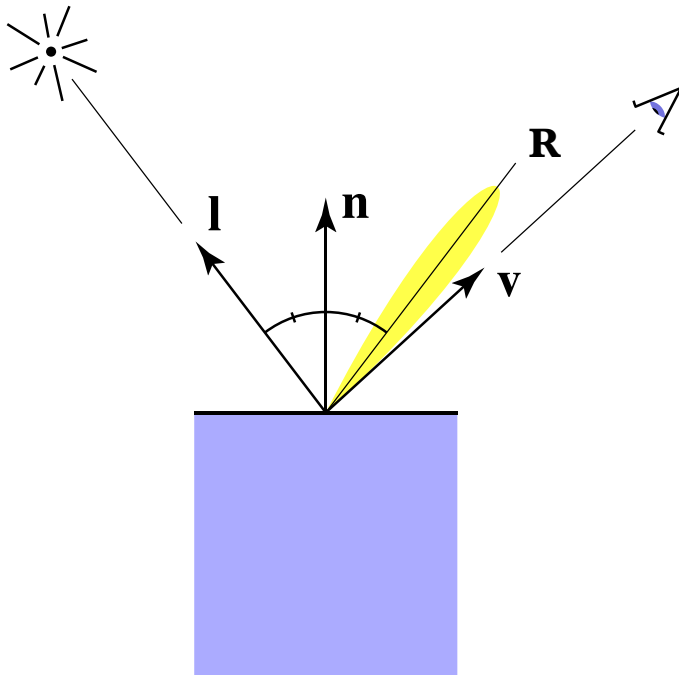


$k_d \longrightarrow$

Specular reflection

Intensity **depends** on view direction

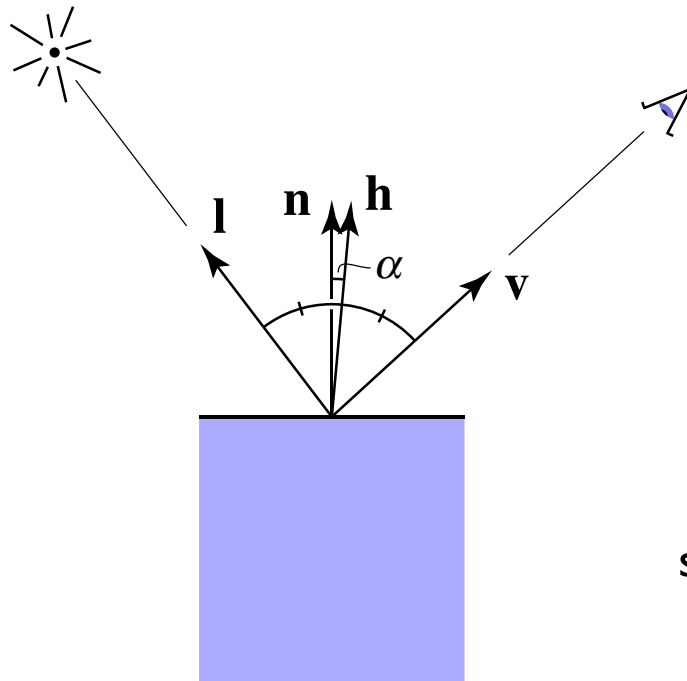
- Bright near mirror reflection direction



Specular reflection

V close to mirror direction $\Leftrightarrow$ **half vector near normal**

- Measure "near" by dot product of unit vectors

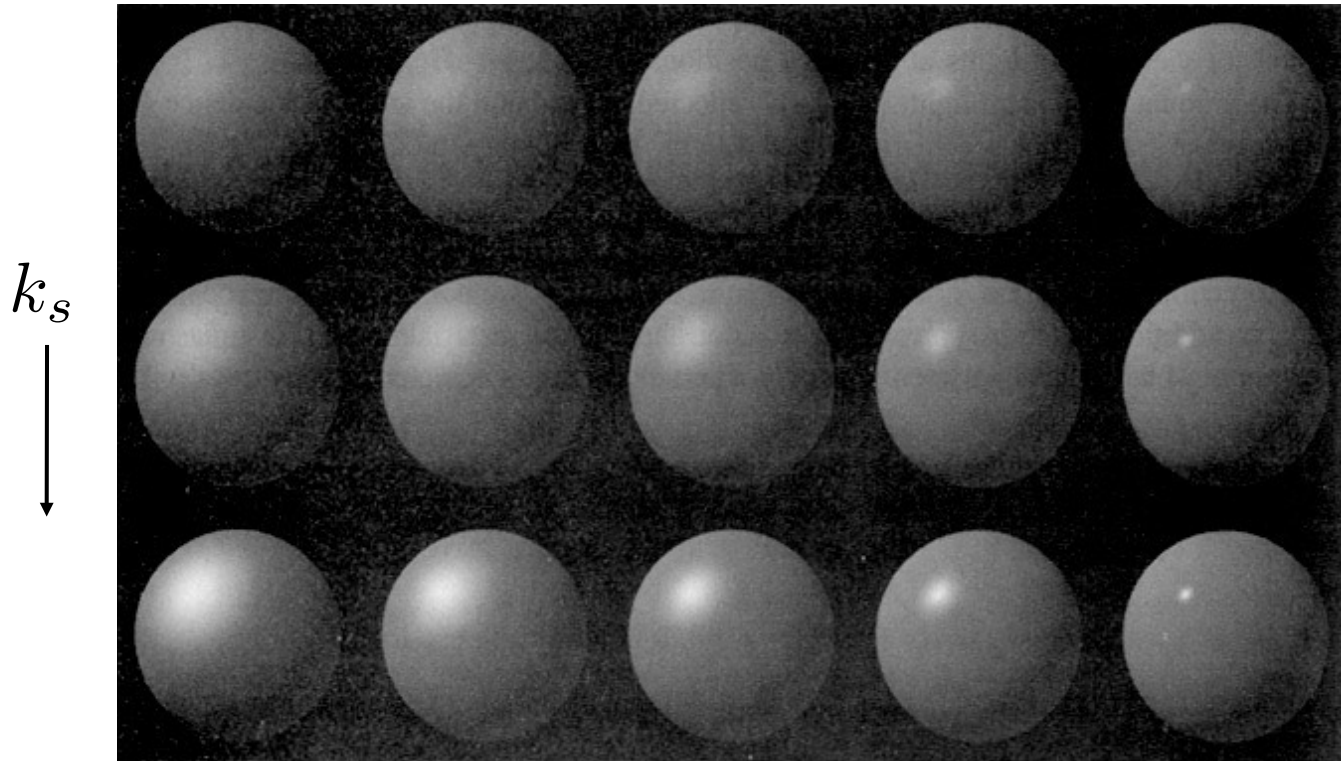


$$\begin{aligned}\mathbf{h} &= \text{bisector}(\mathbf{v}, \mathbf{l}) \\ &\text{(半程向量)} \\ &= \frac{\mathbf{v} + \mathbf{l}}{\|\mathbf{v} + \mathbf{l}\|}\end{aligned}$$

$$\begin{aligned}L_s &= k_s (I/r^2) \max(0, \cos \alpha)^p \\ &= k_s (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p\end{aligned}$$

↑
specularly reflected light ↑
specular coefficient

Specular reflection



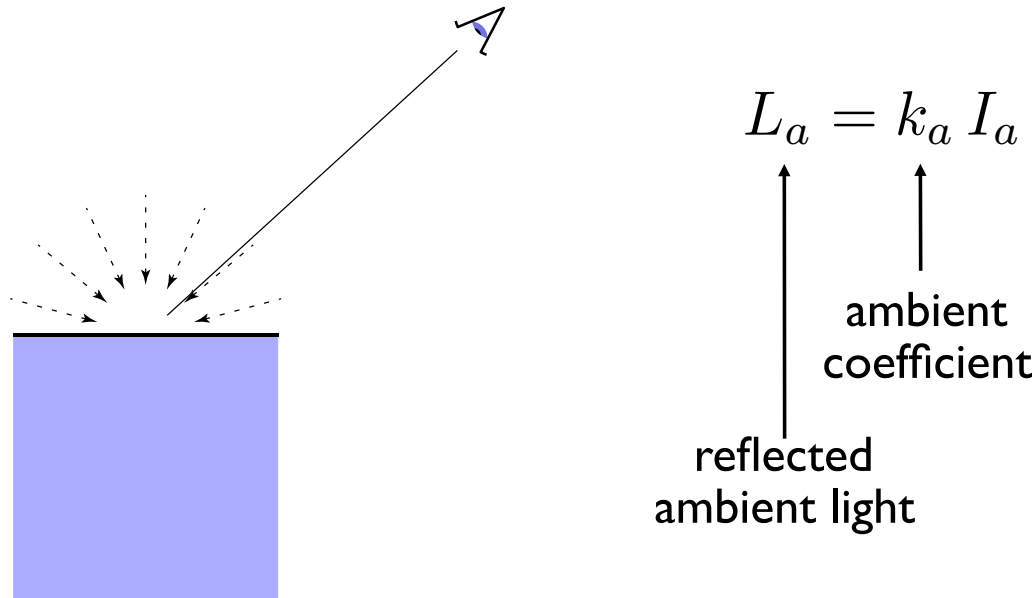
Note: showing
Ld + Ls together

$p \longrightarrow$

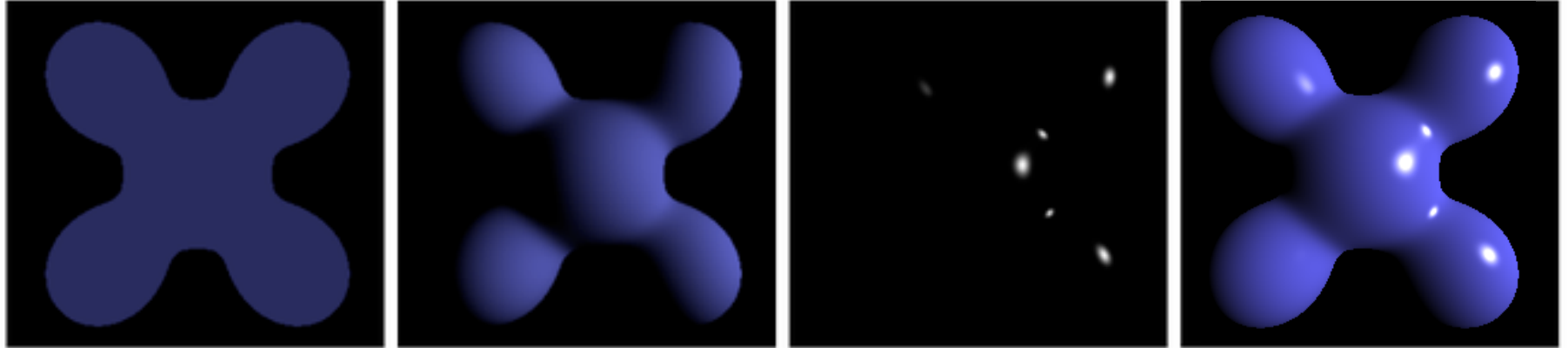
Ambient lighting

Shading that does not depend on anything

- Add constant color to account for disregarded illumination and fill in black shadows
- This is approximate / fake!



Blinn-Phong Reflection Model



Ambient + Diffuse + Specular = Blinn-Phong Reflection

$$\begin{aligned} L &= L_a + L_d + L_s \\ &= k_a I_a + k_d (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{l}) + k_s (I/r^2) \max(0, \mathbf{n} \cdot \mathbf{h})^p \end{aligned}$$

Questions?